

Functional characterisation of neuropeptides that act as ligands for both calcitonin-type and pigment-dispersing factor-type receptors in a deuterostome

Reviewed Preprint

v3 • October 28, 2025

Revised by authors

Reviewed Preprint

v2 • August 21, 2025

Reviewed Preprint

v1 • September 23, 2024

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eLife Assessment

This **valuable** study characterises receptors for calcitonin-related peptides from a deuterostomian animal, the echinoderm *Apostichopus japonicus*, by a combination of heterologous expression, pharmacological experiments, and the quantification of gene-expression levels. The authors provide **convincing** evidence for a functional calcitonin-related peptide system in the sea cucumber, but further work will be needed to confirm the proposed physiological functions of PDF receptor system in this species. This work should be of interest to scientists studying the signaling pathways, functions, and evolution of neuropeptides, and could be of relevance to improving the culture conditions of this economically key species.

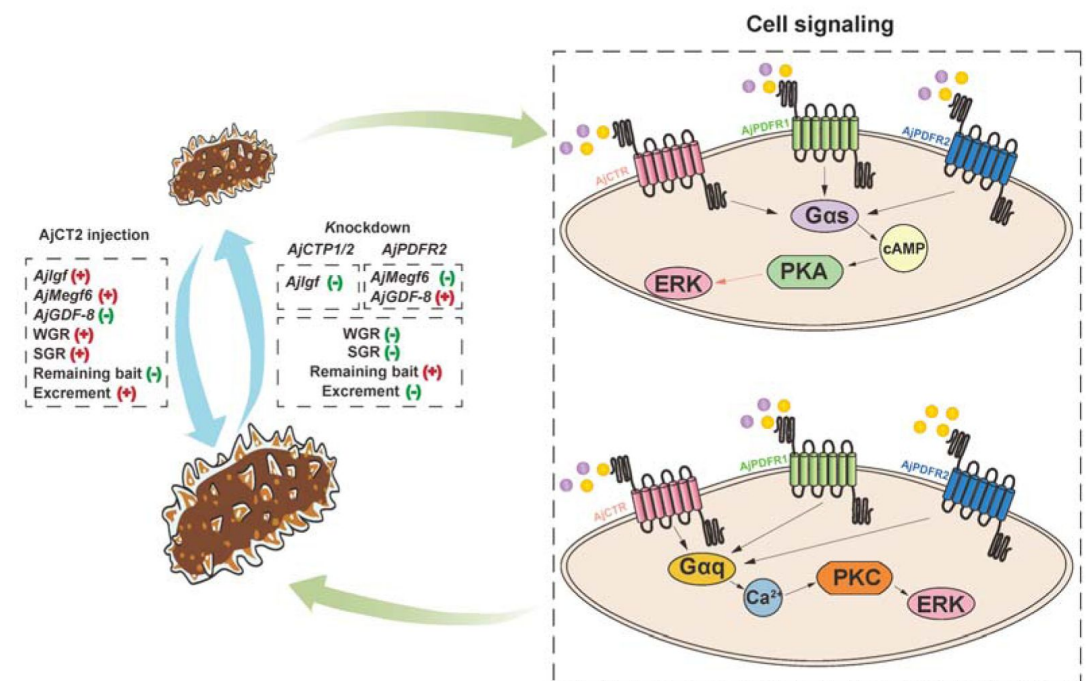
<https://doi.org/10.7554/eLife.101799.3.sa3>

Abstract

The calcitonin (CT) family of related peptides exerts diverse physiological effects in mammals via two G-protein coupled receptors: CTR and the CTR-like receptor CLR. Phylogenetic analysis of CT-type signaling has revealed the presence of CT-type peptides and CTR/CLR-type proteins in both deuterostome and protostome invertebrates. Furthermore, experimental studies have demonstrated that in the protostome *Drosophila melanogaster* the CT-like peptide DH₃₁ can act as a ligand for a CTR/CLR-type receptor and a pigment-dispersing factor (PDF) receptor. Here we investigated the signaling mechanisms and functions of CT-type neuropeptides in a deuterostome invertebrate, the sea cucumber *Apostichopus japonicus* (phylum Echinodermata). In *A. japonicus*, a single gene encodes two CT-type peptides (AjCT1 and AjCT2) and both peptides act as ligands for a CTR/CLR-type receptor (AjCTR) and two PDF-type receptors (AjPDFR1, AjPDFR2), but with differential activation of downstream cAMP/PKA, Gαq/Ca²⁺/PKC and ERK1/2 signaling pathways. AjCT1/AjCT2-encoding transcripts

were detected in the central nervous system and a variety of organ systems and neuropeptide expression was visualized immunohistochemically using an antiserum to a starfish CT-type peptide (ArCT). *In vitro* pharmacological experiments demonstrated that AjCT1 and/or AjCT2 cause dose-dependent relaxation of longitudinal muscle and intestine preparations. Furthermore, *in vivo* pharmacological experiments, combined with gain-and loss-of-function experiments, revealed a potential physiological role for AjCT2/AjPDFR2 signaling in promoting feeding and growth in *A. japonicus*. To our knowledge, this is the first study to obtain evidence that CT-type peptides can act as ligands for both CTR/CLR-type and PDF-type receptors in a deuterostome. Moreover, it provides the first evidence for appetite-stimulating and growth-promoting effects of CT-type neuropeptides in bilaterians. Given the economic importance of *A. japonicus* as a foodstuff, the discovery of CT-type peptides as potential regulators of feeding and growth in this species may offer novel strategies for aquaculture applications.

Graphical abstract



A schematic showing proposed molecular mechanisms by which CT-type signaling regulates feeding and growth in *A. japonicus*. AjCT1 and AjCT2 are represented by purple and yellow circles, respectively. For cell signaling, black arrows indicate that this process is activated, whilst the pink arrow indicates that ERK1/2 is activated in the AjCT1/AjPDFR1, AjCT2/AjPDFR1 and AjCT2/AjPDFR2 signaling pathways.

1. Introduction

The peptide calcitonin (CT) was first identified as a hormone that regulates blood calcium levels and subsequently a family of CT-related neuropeptides was discovered in mammals, including calcitonin gene related peptides (CGRPs), amylin (AMY), adrenomedullin (AM), and CT receptor-stimulating peptide (CRSP) (Copp et al., 1962 [DOI](#); Sekiguchi et al., 2009 [DOI](#); Sekiguchi et al., 2016 [DOI](#)). Members of the CT-type family of peptides typically have a disulfide bond in their N-terminal region (Cai et al., 2018 [DOI](#)) and exert their physiological effects in mammals by binding to two

related G protein-coupled receptors: CTR (CT receptor) and CLR (CTR-like receptor) (Martins et al., 2014). However, receptor activity-modifying proteins (RAMPs) acting as accessory proteins are key determinants of ligand selectivity for CTR and CLR (Mclatchie et al., 1998). Thus, formation of a complex with a RAMP is essential for CLR function, whereas RAMPs are not required for CTR ligand recognition (Kuwasako et al., 2011; Bomberger et al., 2012). Furthermore, binding of CT to CTR in the absence of RAMPs (Katafuchi et al., 2009) can activate signaling via several downstream pathways, including cAMP accumulation, Ca^{2+} mobilization, and ERK activation (Chen et al., 1998; Pondel, 2000; Walker et al., 2010; Andreassen et al., 2014).

Transcriptome/genome sequencing has enabled analysis of the phylogenetic distribution of CT-type signaling systems, indicating that the evolutionary origin of CT-type peptides and CTR/CLR-related receptors can be traced back to the urbilaterian common ancestor of deuterostomes and protostomes (Furuya et al., 2000; Johnson et al., 2005; Jékely, 2013; Mirabeau and Joly, 2013). Furthermore, in protostomes two different types of CT-related peptides have been identified. Firstly, peptides that are structurally similar to vertebrate CT-type peptides in having an N-terminal disulphide bridge and secondly, CT-like diuretic hormone 31 (DH₃₁)-type peptides that lack an N-terminal disulphide bridge (Furuya et al., 2000; Conzelmann et al., 2013). Furthermore, phylogenetic analysis indicates that gene duplication in a common ancestor of the protostomes gave rise to genes encoding CT-type peptides with a disulphide bridge and DH₃₁-type peptides, which were then differentially retained in protostome taxa (Conzelmann et al., 2013; Cai et al., 2018).

Receptors for CT-type peptides with a disulphide bridge have been characterised in a molluscan species, the oyster *Crassostrea gigas* (now known as *Magallana gigas*), which has seven CTR/CLR-type receptors. Two of these receptors, named Cg-CT-R and Cragi-CTR2, were shown to be specifically activated by the CT-type neuropeptides Cragi-CT1b and Cragi-CT2, respectively, thereby mediating intracellular Ca^{2+} mobilization and cAMP accumulation (Schwartz et al., 2019). CT-type peptides with a disulphide bridge are not present in the insect *Drosophila melanogaster*. However, receptors that are activated by DH₃₁ have been identified in this species and, interestingly, DH₃₁ can act as a ligand for two types of receptors. Thus, DH₃₁ acts as a ligand for a CTR/CLR-related protein (CG17415), but it also acts as a ligand for a protein (CG13758) that is the receptor for pigment-dispersing factor (PDF) (Johnson et al., 2005; Mertens et al., 2005; Shafer et al., 2008). Furthermore, the functional significance of the existence of dual receptor signaling pathways for DH₃₁ in *Drosophila* has been investigated extensively (Johnson et al., 2005; Goda et al., 2019).

Currently, relatively little is known about CT-type signaling systems in deuterostome invertebrates. Analysis of transcriptome/genome sequence data has enabled identification of genes/transcripts encoding CT-type peptides and candidate CTR/CLR-type receptors for these peptides in several deuterostome invertebrate taxa, including urochordates (e.g. *Ciona intestinalis*), cephalochordates (e.g. *Branchiostoma floridae*), hemichordates (e.g. *Saccoglossus kowalevskii*) and echinoderms (e.g. *Strongylocentrotus purpuratus*) (Burke et al., 2006; Sekiguchi et al., 2009; Sekiguchi et al., 2016; Rowe and Elphick, 2012; Martins et al., 2014). Furthermore, analysis of the expression and action of CT-type peptides identified in the starfish *Asterias rubens* and *Patiria pectinifera* has revealed that they act as muscle relaxants (Cai et al., 2018; Semmens et al., 2016). However, experimental confirmation of ligand-receptor partners has thus far only been reported for the cephalochordate *B. floridae*, where three CT-type peptides were shown to act as ligands for a CTR/CLR-type receptor when it is co-expressed with RAMP-type proteins (Sekiguchi et al., 2016). To gain further insights into the mechanisms of CT-type signaling in invertebrate deuterostomes, the aim of this study was to functionally characterise CT-type peptides and their receptors in an echinoderm – the sea cucumber *Apostichopus japonicus*.

A. japonicus is an economically important species in Asian aquaculture (Pangestuti and Arifin, 2018 [↗](#); Guo et al., 2020 [↗](#)) and a species representative of the phylum Echinodermata. As a phylum of deuterostome invertebrates, echinoderms form a sister clade to the chordates together with hemichordates and xenoturbellids (Bromham and Degnan, 1999 [↗](#); Bourlat et al., 2006 [↗](#); Dunn et al., 2008 [↗](#)) and as such they occupy a key “intermediate” phylogenetic position relative to the protostome invertebrates and vertebrates (Zandawala et al., 2017 [↗](#)), which make them attractive as experimental systems to address evolutionary questions. In our previous studies, we have established this species as a model system to obtain insights into the evolutionary history and comparative physiology of neuropeptide signaling systems in bilaterians (Chen et al., 2019 [↗](#); Li et al., 2022 [↗](#); Zheng et al., 2022 [↗](#); Cong et al., 2023 [↗](#); Zheng et al., 2024 [↗](#)). A gene encoding two CT-type peptides (AjCT1 and AjCT2) has been identified in *A. japonicus* (Chen et al., 2019 [↗](#)) and a candidate receptor (AjCTR) for these peptides has also been predicted (Huang et al., 2021 [↗](#)). Furthermore, relevant to the discovery that the CT-like peptide DH₃₁ can act as a ligand for both a CTR/CLR-type receptor and a PDF-type receptor in *Drosophila*, the sequences of two PDF-type receptors have been identified in *A. japonicus* (AjPDFR1 and AjPDFR2) (Huang et al., 2021 [↗](#)).

The first aim of this study was to investigate if AjCT1 and/or AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2 and to characterise the downstream signaling pathways that are activated when AjCT1 and/or AjCT2 bind to their receptor(s). The second aim was to investigate the physiological roles of AjCT1 and AjCT2 in *A. japonicus* by analysing the expression of the gene encoding these peptides, investigating CT-type neuropeptide expression using immunohistochemistry, examining the *in vitro* and *in vivo* pharmacological effects of these peptides and performing loss-of-function tests. The functional characterisation of CT-type signaling in *A. japonicus* reported in this study provides new insights into the evolution and comparative physiology of neuropeptides and a basis for application of neuropeptide-based strategies to improve aquaculture methods for this economically important species.

2. Results

2.1 Characterization and phylogenetic analysis of AjCT1/2

The complete cDNA sequences of the CT-type neuropeptide precursors *AjCTP1* and *AjCTP2* were confirmed. The coding sequence of *AjCTP2* is shorter than *AjCTP1* (Figure 1A [↗](#)) but both transcripts encode precursor proteins that comprise the same predicted 23-residue N-terminal signal peptide (Figure 1-figure supplement 1A [↗](#)). *AjCTP1* encodes both putative CT-type neuropeptides (AjCT1 and AjCT2), while *AjCTP2* only encodes AjCT2 (Figure 1A [↗](#), Figure 1-figure supplement 1A [↗](#) and 1B [↗](#)). Exon and intron structure analysis showed that *AjCTP1* comprises four exons, with AjCT1 and AjCT2 encoded in the third and fourth exons, respectively. *AjCTP2* comprises three exons, excluding the third exon that encodes AjCT1 (Figure 1A [↗](#)). Phylogenetic analysis revealed that *AjCTP1* and *AjCTP2* are closely related to CT-type precursors from other sea cucumbers (*Holothuria glaberrima* and *Holothuria scabra*), and are positioned within a clade that also contains CT-type precursors from other echinoderms (sea urchin *S. purpuratus*, starfish *A. rubens*). Furthermore, the echinoderm CT-type precursors are positioned within a clade that also comprises CT-type precursors from chordates, whilst DH₃₁/CT-type precursors from protostomes are positioned in a separate clade (Figure 1B [↗](#)). Sequence alignment of CT-related neuropeptides revealed that, with the exception of DH₃₁-type peptides, the N-terminal region of CT-related peptides has a pair of conserved cysteine residues that are known or predicted to form a disulfide bond. Another conserved feature is a C-terminal proline residue, with the exception of CGRPs (Figure 1-figure supplement 2 [↗](#)). Accordingly, both AjCT1 and AjCT2 have these conserved characteristics.

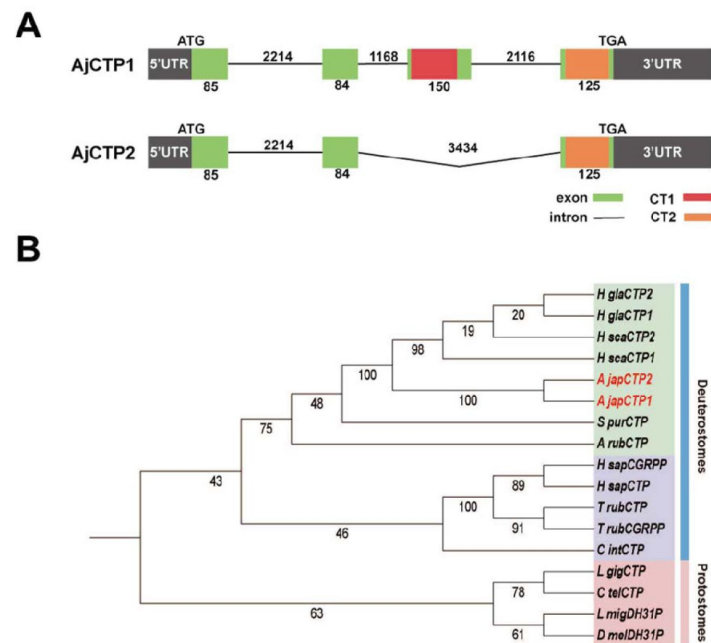


Figure 1.

CT-type neuropeptide precursors and candidate receptors for CT-type peptides in *A. japonicus*.

(A) Alternative splicing schematic of *AjCTP1* and *AjCTP2*. Exons are represented by green rectangles, introns are represented by lines, *AjCT1* and *AjCT2* are represented by red and orange rectangles respectively, upstream and downstream non-coding regions are represented by gray rectangles. The numbers represent the length of exons or introns. (B) Phylogenetic analysis of *AjCTP1* and *AjCTP2* and CT-related neuropeptide precursors from other bilaterians. CT-type precursors from deuterostomes are shown in green (echinoderms) and purple (chordates), CT-type precursors from protostomes are shown in pink. Full species names and accession numbers are listed in [Supplementary table 1](#).

2.2 Characterization and phylogenetic analysis of three candidate receptors for AjCT1/2

The sequences of three candidate receptors for AjCT1 and AjCT2 were obtained by RACE and phosphorylation sites were predicted, as shown in **Figure 2-figure supplement 1A**, **1B**, and **1C**. Phylogenetic analysis revealed that CTR/CLR-related receptors divided into three clades and an *A. japonicus* CTR/CLR-type receptor (AjCTR) was positioned in a clade with other deuterostome CTR/CLR-type receptors, including human CTR and CLR. The other two clades comprised protostome DH31-type receptors, which include the *Drosophila* receptor that has been identified experimentally as the receptor for DH31, and other protostome CTR/CLR-related receptors (Cluster A), including two receptors in the mollusc *Magallana gigas* that have been identified experimentally as receptors for CT-like peptides with a disulphide bridge. The phylogenetic analysis also revealed that two *A. japonicus* PDF-type receptors (AjPDFR1 and AjPDFR2) were positioned in a clade comprising PDF-type receptors, with one branch containing PDF-type receptors from deuterostomes (including AjPDFR1 and AjPDFR2) and another branch containing PDF-type receptors from protostomes (including the *Drosophila* PDF receptor) (**Figure 2**). Domain prediction of the candidate receptors showed that AjCTR had HRM (Hormone receptor) and 7tm_GPCRs (seven-transmembrane G protein-coupled receptor) superfamily domains (**Figure 2-figure supplement 2A**), whereas AjPDFR1 and AjPDFR2 only presented the 7tm_GPCRs superfamily domain (**Figure 2-figure supplement 2B**). To further investigate an orthologous relationship of the candidate receptors with homologous receptors in other echinoderms, we conducted a gene synteny analysis in *A. japonicus*, *A. rubens* and *L. variegatus*. The neighboring genes of AjCTR included *GOLGA4*, *NeuroD*, *CRF2*, *ppGalNAc-T*, *URB3*, *SLC40A1*, *CMAH* and *MAGP*. The relative positions of these genes were different from those in *A. rubens* and *L. variegatus* (**Figure 2-figure supplement 2C**). The same situation was also observed for the synteny analysis of PDFRs.

2.3 AjCT1 and AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2

To investigate if AjCT1 and/or AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2, here, two receptor assay methods were employed. As shown in **Figure 3A**, both AjCT1 and AjCT2 (10^{-6} M) caused a significant increase in CRE luciferase activity in HEK293T cells transfected with AjCTR, AjPDFR1 or AjPDFR2. Furthermore, this effect was inhibited by pre-incubation of cells with the PKA inhibitor H89. The intracellular accumulation of cAMP was also measured, and this showed that AjCT1 and AjCT2 (10^{-9} – 10^{-5} M) caused a dose-dependent increase in cAMP in HEK293T cells transfected with AjCTR, AjPDFR1 or AjPDFR2 (**Figure 3B**). However, when exposed to AjCT1 or AjCT2, the maximum accumulation of intracellular cAMP was higher in cells transfected with AjCTR than in cells transfected with AjPDFR1 and AjPDFR2 (**Figure 3B**). Measurement of SRE luciferase activity revealed that AjCT1 and AjCT2 (10^{-6} M) administration evoked an increase in activity in HEK293T cells transfected with AjCTR and AjPDFR1 (**Figure 4A**). However, in HEK293T cells transfected with AjPDFR2, an increase in SRE luciferase activity was only observed with AjCT2, not with AjCT1. Furthermore, as shown in **Figure 4A**, AjCT1- and AjCT2-induced increases in SRE luciferase activity were not observed in cells that had been pre-incubated with FR900359 (Gaq protein inhibitor) and Gö 6983 (PKC inhibitor). As shown in **Figure 4-figure supplement 1**, receptors localized in the cell membrane, but after exposure to AjCT1 and AjCT2 (10^{-6} M) for 15 min, internalization of pEGFP-N1/AjCTR, pEGFP-N1/AjPDFR1 and pEGFP-N1/AjPDFR2 was observed in transfected HEK293T cells (**Figure 4B**). Taken together, all these findings indicate that AjCT1 and AjCT2 can act as ligands for the three candidate receptors and then activate cAMP/PKA signaling and Gaq/Ca²⁺/PKC signaling in all combinations, with the exception of AjCT1/AjPDFR2 which only activates the cAMP/PKA cascade.

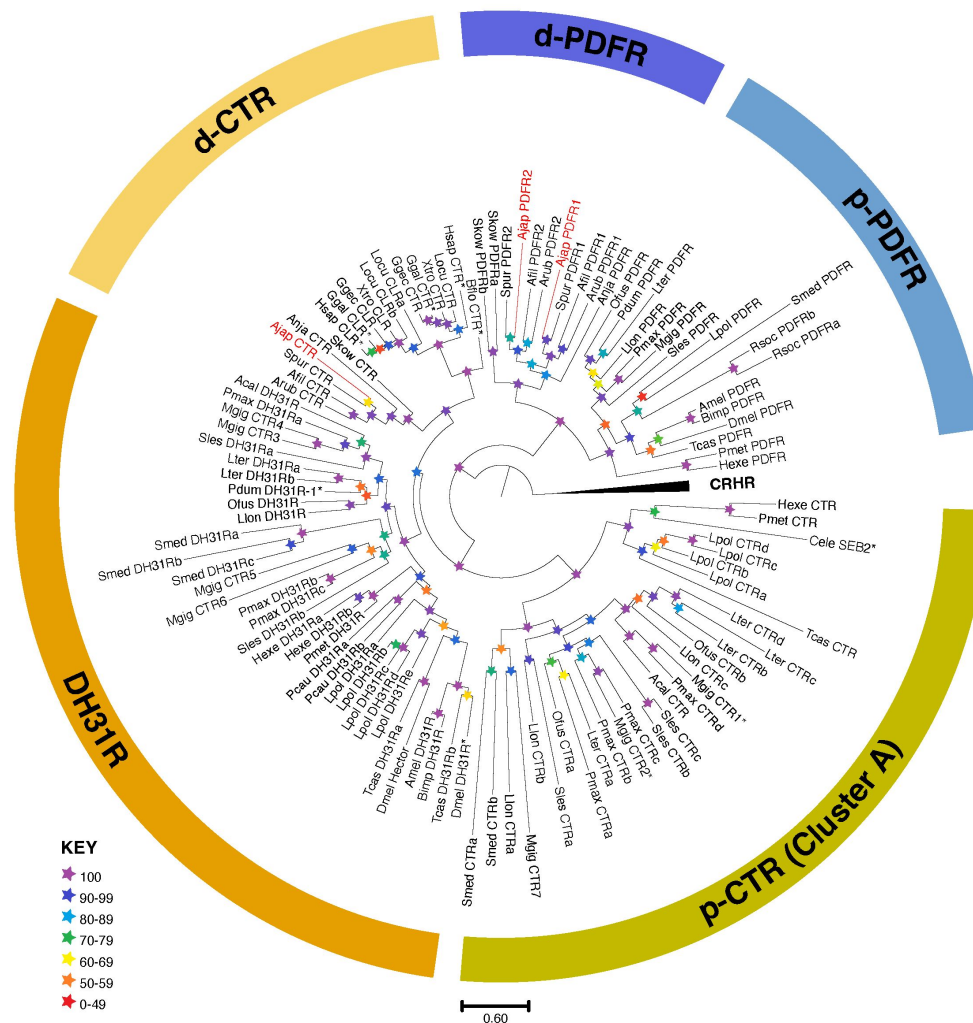


Figure 2.

Maximum-likelihood phylogenetic tree of bilaterian CT-type and PDF-type receptor families, including those found in *A. japonicus*, rooted with CRH-type receptors.

The tree comprises distinct CT-type receptor and PDF-type receptor clades with bootstrap support >90. The CT-type receptor clade is subdivided into a deuterostome-specific clade (dCTR, yellow), which includes the *A. japonicus* CTR (AjCTR), and two protostome receptor clades. One comprises DH31 receptors (DH31R, orange), and the other comprises protostome-specific CTRs (p-CTR (Cluster A), green). The PDF-type receptors are subdivided into protostome (p-PDFR, blue) and deuterostome (d-PDFR, purple) clades, with the latter including the two *A. japonicus* PDFRs (AjPDFR1, AjPDFR2). The stars represent branch support (bootstrap 1000 replicates, see key). The scale bar indicates amino acid substitutions per site. The *A. japonicus* receptors characterised in this paper are highlighted in red, and CT-type family receptors for which ligands have been identified experimentally in other published studies are indicated with an asterisk (Gorn et al., 1992; McLatchie et al., 1998; Johnson et al., 2005; Bauknecht & Jekely, 2015; Sekiguchi et al., 2016; Schwartz et al., 2019; Beets et al., 2023; Huang et al., 2024). Species names are abbreviated as follows: Acal (*Aplysia californica*), Afil (*Amphiura filiformis*), Ajap (*Apostichopus japonicus*), Amel (*Apis mellifera*), Anja (*Anneissia japonica*), Arub (*Asterias rubens*), Bflo (*Branchiostoma floridae*), Bimp (*Bombus impatiens*), Cele (*Caenorhabditis elegans*), Cint (*Ciona intestinalis*), Dmel (*Drosophila melanogaster*), Ggal (*Gallus gallus*), Ggek (*Gekko gekko*), Hexe (*Hypsibius exemplaris*), Hsap (*Homo sapiens*), Llon (*Lineus longissimus*), Locu (*Lepisosteus oculatus*), Lpol (*Limulus polyphemus*), Lter (*Lumbricus terrestris*), Mgig (*Magallana gigas*), Ofus (*Owenia fusiformis*), Pcaw (*Priapulus caudatus*), Pdum (*Platynereis dumerilii*), Pmax (*Pecten maximus*), Pmet (*Paramacrobrotus metropolita*), Rsoc (*Rotaria socialis*), Scla (*Styela clava*), Skow (*Saccoglossus kowalevskii*), Sles (*Sepioteuthis laessoniana*), Smed (*Schmidtea mediterranea*), Spur (*Strongylocentrotus purpuratus*), Tcas (*Tribolium castaneum*), Xtro (*Xenopus tropicalis*). Sequence accession numbers are listed in [supplementary table 2](#).

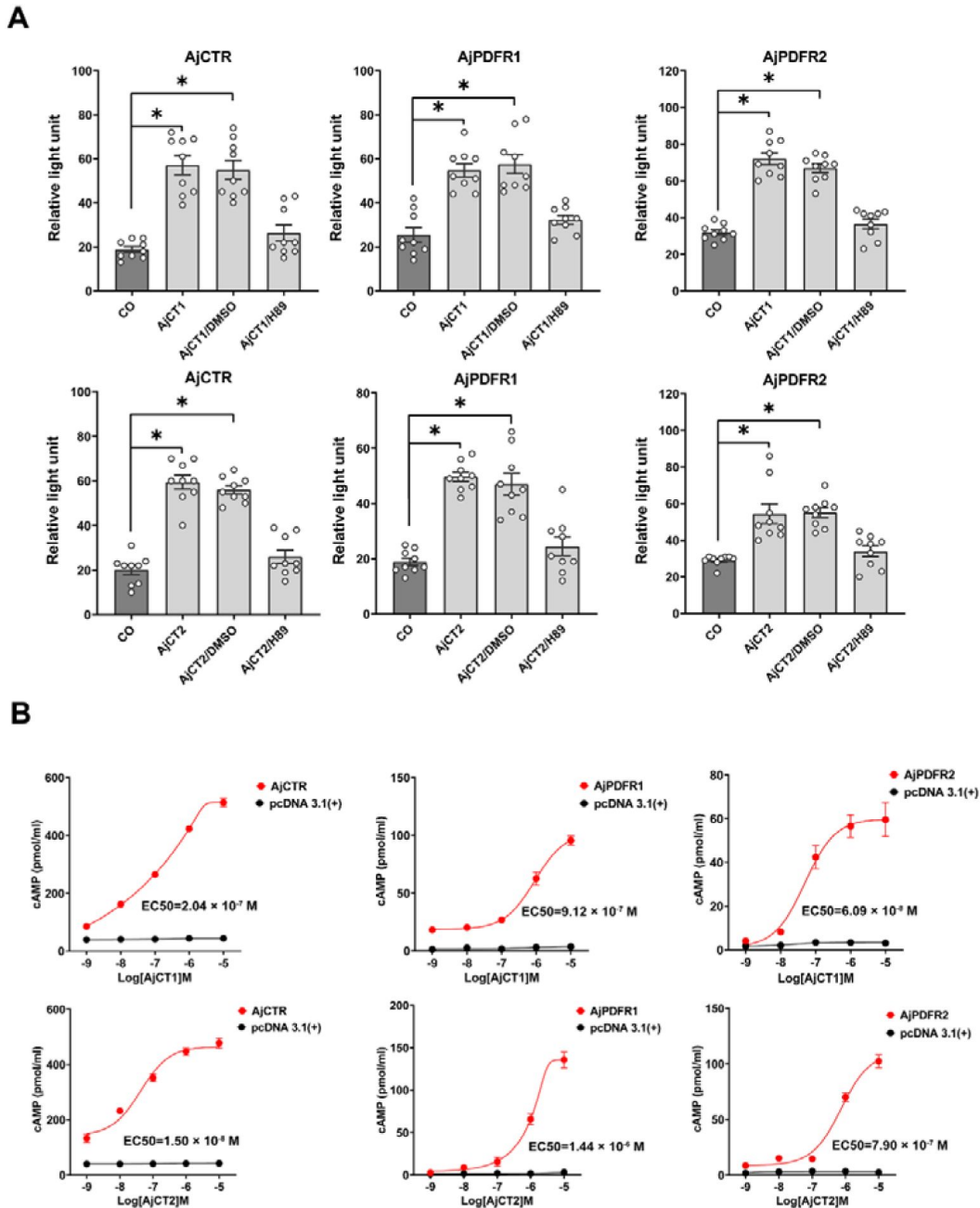
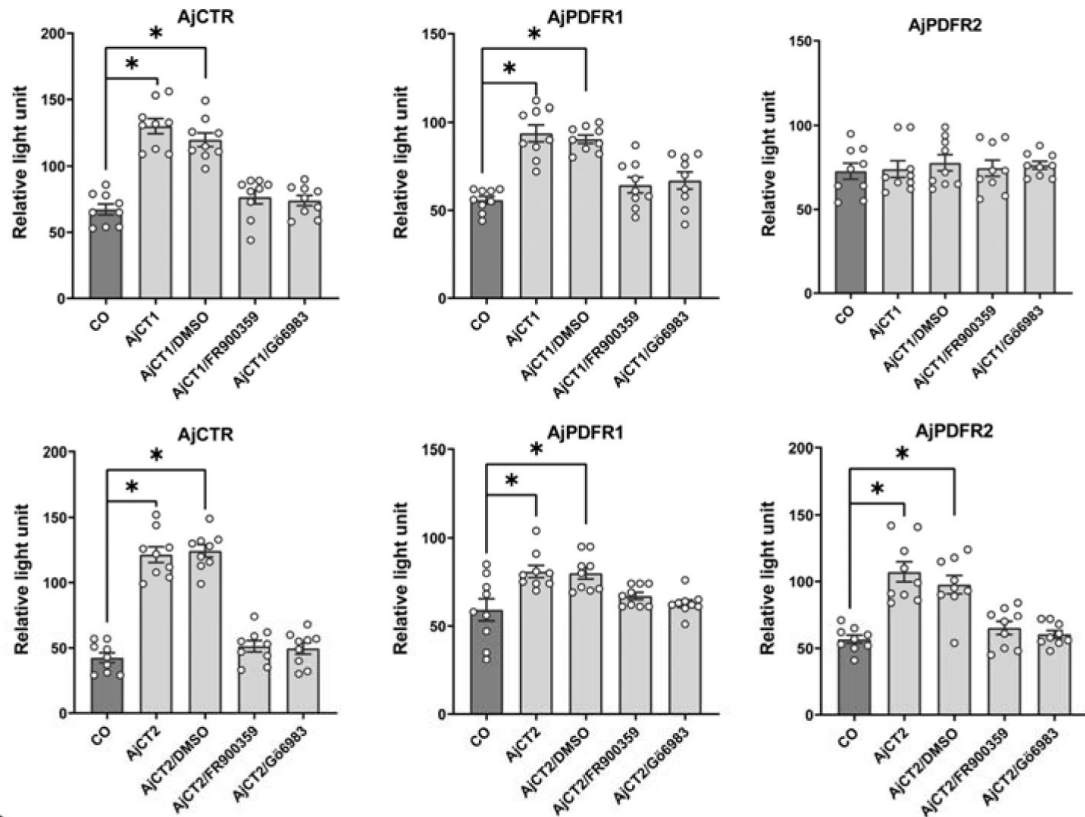


Figure 3.

Pharmacological characterisation of AjCT1 and AjCT2 as ligands for the *A. japonicus* receptors AjCTR, AjPDFR1 and AjPDFR2.

(A) CRE-driven luciferase activity measured in HEK293T cells transfected with one of the three candidate receptors after exposure to AjCT1 or AjCT2 or neuropeptide + DMSO or neuropeptide + H89 (10 μ M) or serum-free DMEM (CO, negative control). The neuropeptide concentration used here was 10⁻⁶ M. Mean values with standard deviations (n = 9) are shown. * indicates a statistically significant difference with $p < 0.05$. (B) Measurement of cAMP accumulation after exposure to AjCT1 or AjCT2 (10⁻⁹ – 10⁻⁵ M). No cAMP elevation is observed in control experiments where HEK293T cells were transfected with empty pcDNA 3.1(+) (black circle). The estimated EC₅₀ of AjCT1 for AjCTR, AjPDFR1 and AjPDFR2 were 2.04 × 10⁻⁷ M (95% CI: 1.06 × 10⁻⁷ – 8.69 × 10⁻⁷ M), 9.12 × 10⁻⁷ M (95% CI: 5.56 × 10⁻⁷ – 4.45 × 10⁻⁶ M) and 6.09 × 10⁻⁸ M (95% CI: 2.23 × 10⁻⁸ – 2.84 × 10⁻⁷ M) respectively. The estimated EC₅₀ of AjCT2 for AjCTR, AjPDFR1 and AjPDFR2 were 1.50 × 10⁻⁸ M (95% CI: 1.03 × 10⁻⁹ – 3.89 × 10⁻⁸ M), 1.44 × 10⁻⁶ M (95% CI: 1.04 × 10⁻⁶ – 1.99 × 10⁻⁶ M) and 7.90 × 10⁻⁷ M (95% CI: 5.73 × 10⁻⁷ – 1.06 × 10⁻⁶ M) respectively. Error bars represent SEM for three independent experiments.

A



B

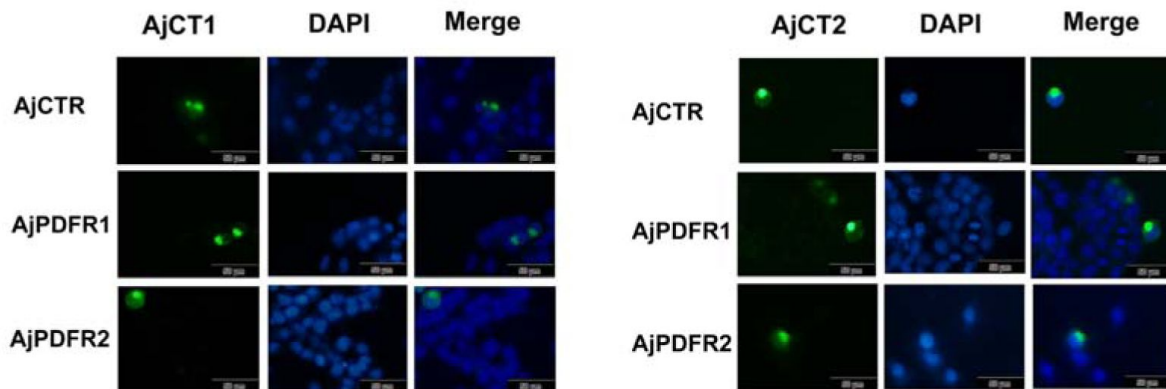


Figure 4.

SRE-driven luciferase activity and receptor internalization in cells transfected with AjCTR, AjPDFR1 or AjPDFR2 and exposed to AjCT1 or AjCT2.

(A) SRE-driven luciferase activity measured after incubation with neuropeptide (10^{-6} M AjCT1 or AjCT2)/DMSO, neuropeptide (10^{-6} M)/FR900359 (1 μ M) and neuropeptide (10^{-6} M)/Gö 6983 (1 μ M). Mean values with standard deviations ($n = 9$) are shown. * indicates a statistically significant difference with $p < 0.05$. (B) Internalization of AjCTR, AjPDFR1 and AjPDFR2 after 15 min treatment with 10^{-6} M AjCT1 or AjCT2. Green fluorescence represents the localization of pEGFP-N1/receptors in cells. The cell nucleus probe (DAPI) was used for cell nucleus staining. Scale bar: 50 μ m. All pictures are representative of three independent experiments.

To investigate whether the activated signaling pathways may exert downstream effects via the ERK–MAPK pathway, the phosphorylation levels of ERK1/2 were analysed in transfected HEK293T cells. pERK1/2 levels in cells transfected with AjCTR or AjPDFR1, but not AjPDFR2, were significantly increased after a 5 min incubation with AjCT1 (**Figure 5A**). Furthermore, this effect was blocked by the PKC inhibitor in AjCTR-expressing cells, and this effect was blocked by both PKA and PKC inhibitors in AjPDFR1-expressing cells (**Figure 5-figure supplement 1A**). pERK1/2 levels in cells transfected with AjCTR or AjPDFR1 or AjPDFR2, were significantly increased after a 5 min incubation with AjCT2 (**Figure 5B**). Moreover, this effect persisted until 15 min after exposure to AjCT in both AjCTR-transfected cells and AjPDFR1-transfected cells but not in AjPDFR2-transfected cells (**Figure 5B**, **Table 1**). Furthermore, these effects were blocked by PKA and PKC inhibitors in AjPDFR1 and AjPDFR2 expressing cells, whereas in AjCTR expressing cells, peptide-induced pERK1/2 activity was only blocked by a PKC inhibitor (**Figure 5-figure supplement 1B**). These results indicate that, with the exception of AjCT1/AjPDFR2, both AjCT1 and AjCT2 can activate the ERK1/2 cascade by binding to the candidate receptors.

2.4 Expression profile of AjCTP1/2 and its receptors in *A. japonicus*

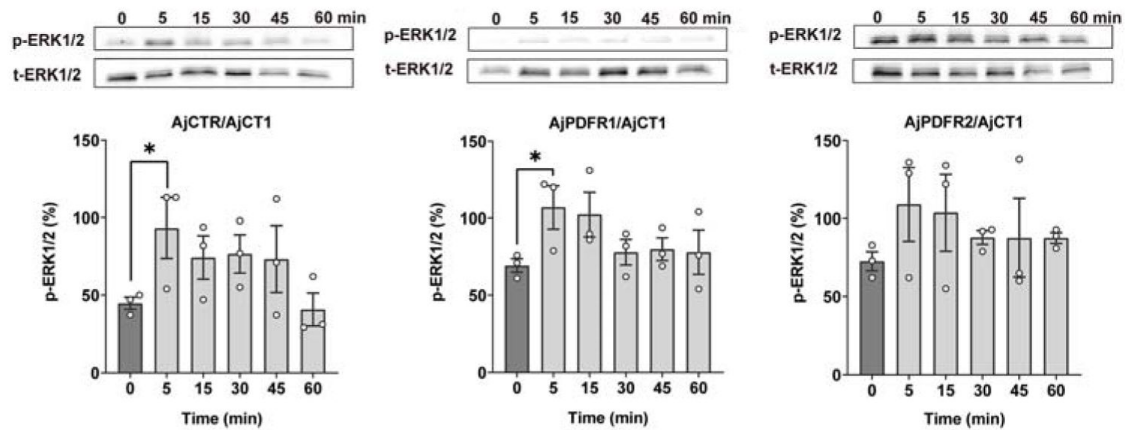
qRT-PCR analysis revealed that *AjCTP1/2* had the highest expression level in the circumoral nervous system (CNS) (**Figure 6A**). The expression levels of three receptors differed from those of *AjCTP1/2*. *AjCTR* (**Figure 6B**) and *AjPDFR1* (**Figure 6C**) showed the highest expression level in CNS and longitudinal muscle, whereas *AjPDFR2* exhibited highest expression in the respiratory tree (**Figure 6D**).

To investigate the distribution of CT-type neuropeptides in *A. japonicus* using immunohistochemistry we employed an antiserum to the starfish (*Asterias rubens*) CT-type neuropeptide ArCT, which has been used previously for visualization of the expression of ArCT in *A. rubens* (Cai et al., 2018) and a CT-type neuropeptide in the feather star *Antedon mediterranea* (AmCT; Aleotti et al., 2022). The antigen peptide that was used to generate the ArCT antiserum (KYNSPFGASGP-NH₂) shares a C-terminal FGxxGP-NH₂ (where X is variable) motif with both AjCT1 (FGSAGP-NH₂) and AjCT2 (FGSGGP-NH₂) and therefore antibodies to ArCT may cross-react with AjCT1 and/or AjCT2. Accordingly, the ArCT antiserum revealed immunostaining in the central nervous system (radial nerve cords and circumoral nerve ring) (**Figure 7A–F**), which was abolished by pre-incubation of the antiserum with the ArCT antigen peptide (**Figure 7-figure supplement 1**). More specifically, high magnification images revealed immunostaining was localised in the ectoneural region, but not in the hyponeural region, of the radial nerve cords, with bilaterally symmetrical immunostained cells (neuronal somata) present in the ectoneural epithelium (**Figure 7C**). Likewise, immunostained cells were revealed in the ectoneural region of the circumoral nerve ring (**Figure 7D–F**). Furthermore, immunostaining was also revealed in tentacle nerves, which emanate from the circumoral nerve ring (**Figure 7D–F**), and in several regions of the digestive system, including the pharynx/stomach (**Figure 7G**), intestine (**Figure 7H**) and cloaca (**Figure 7I**). No immunostaining was observed in the longitudinal muscles (**Figure 7A**, **B**).

2.5 AjCT1 and AjCT2 causes dose-dependent relaxation of longitudinal muscle and intestine preparations from *A. japonicus*

Both AjCT1 and AjCT2 caused dose-dependent relaxation of longitudinal muscles from *A. japonicus* at 10⁻¹⁰–10⁻⁶ M (**Figure 8**). The maximum relaxant effects of AjCT2 (10⁻⁶ M) on longitudinal muscles are illustrated in **Figure 8-figure supplement 1A**. At the highest concentration tested (10⁻⁶ M), AjCT1 and AjCT2 caused 30.07 ± 7.49% and 32.50 ± 14.49% reversal of acetylcholine (ACh)-induced contraction, respectively (**Figure 8A**, **8B**). AjCT1 had no effect on the contractile state of intestine preparations, even at concentrations as high as 10⁻⁵ M (**Figure 8C**). Unlike AjCT1, AjCT2 caused dose-dependent relaxation of intestine preparations, with 26.33 ± 8.75% reversal of

A



B

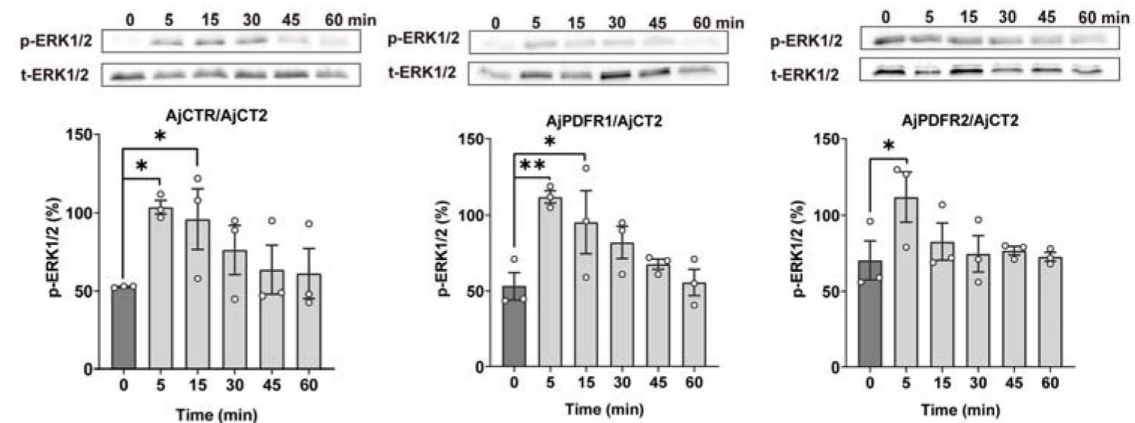


Figure 5.

ERK1/2 activity in cells transfected with AjCTR, AjPDFR1 or AjPDFR2 and exposed to AjCT1 or AjCT2.

(A) and (B) show the immunoblot intensity of representative phosphorylation bands. The transfected HEK293T cells were incubated with AjCT1 (10^{-6} M) or AjCT2 (10^{-6} M) for 0, 5, 15, 30, 45 and 60 min. The p-ERK1/2 was normalized based on t-ERK1/2. All images are representative of three independent experiments. Error bars represent SEM for three independent experiments. Statistically significant differences are indicated as follows: *: $p < 0.05$, **: $p < 0.01$.

Assays Receptors	cAMP/PKA/ERK1/2		Gαq/Ca ²⁺ /PKC/ERK1/2	
	AjCT1	AjCT2	AjCT1	AjCT2
<i>AjCTR</i>	×*	×*	√	√
<i>AjPDR1</i>	√	√	√	√
<i>AjPDR2</i>	×*	√	×	√

Table 1.

Activation of signaling pathways A tick indicates cAMP/PKA/ERK1/2 or Gαq/Ca²⁺/PKC/ERK1/2 activation, the cross indicates no activation, the cross with * indicates that the cAMP/PKA pathway is activated, while the ERK1/2 pathway is not activated.

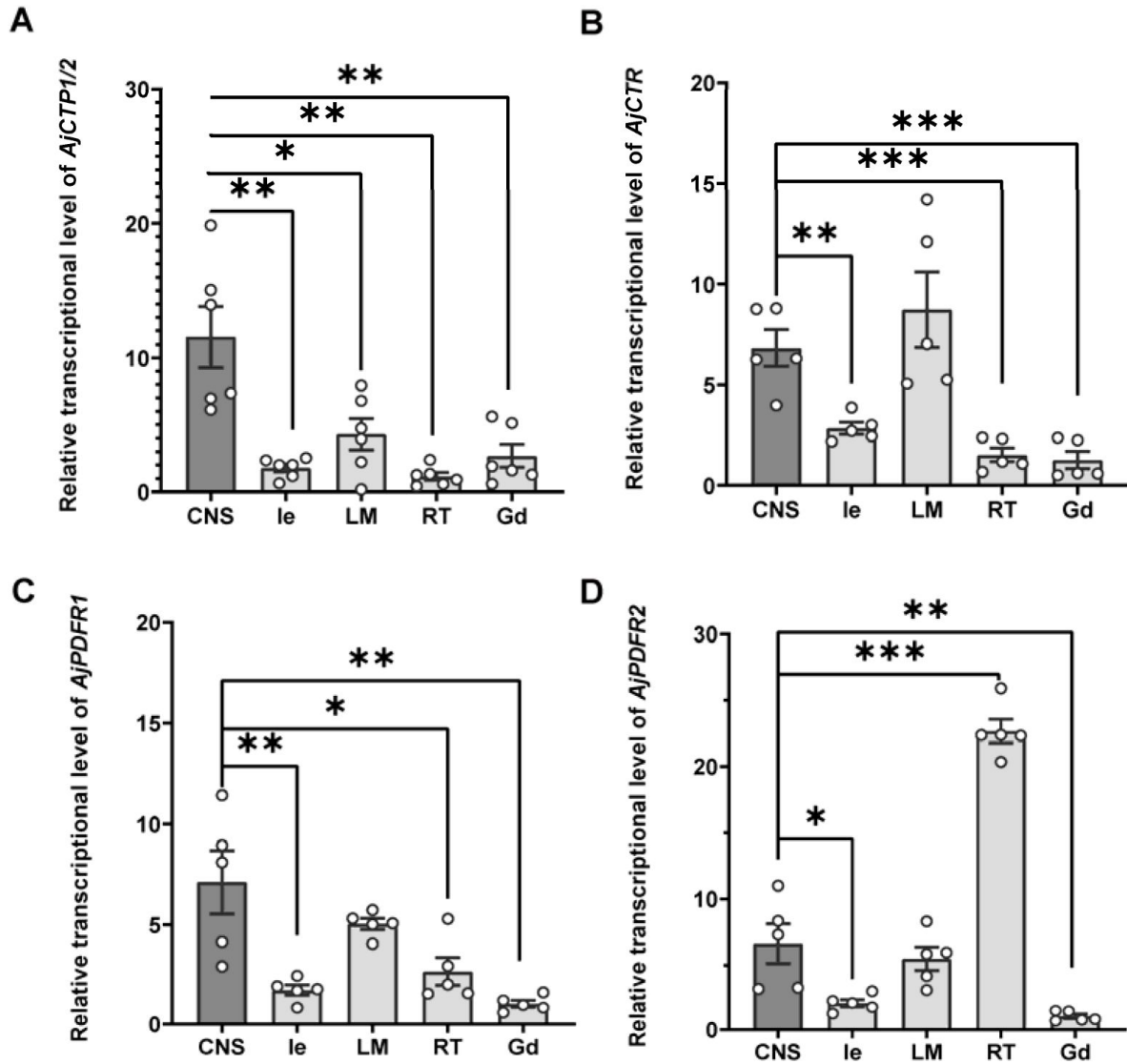


Figure 6.

The relative expression level of *AjCTP1/2* (A), *AjCTR* (B), *AjPDRF1* (C) and *AjPDRF2* (D) in different tissues.

The circumoral nervous system was selected as the control group. Values are means $\pm$ SEM (*AjCTP1/2*, $n = 6$; *AjCTR*, *AjPDRF1* and *AjPDRF2*, $n = 5$). Asterisks represent significant differences as follows: *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$. CNS, circumoral nervous system; Gd, gonad; le, intestine; LM, longitudinal muscle; RT, respiratory tree.

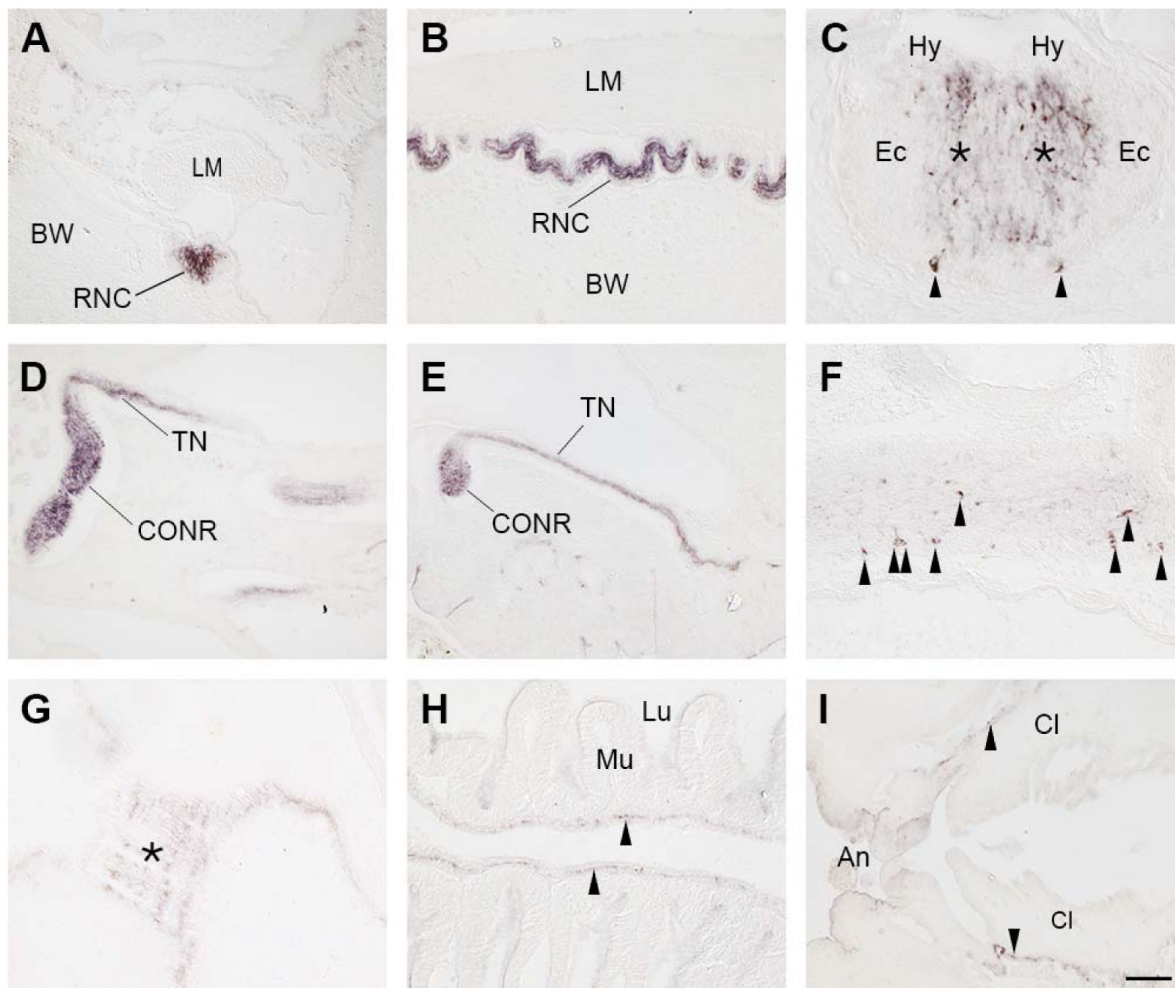


Figure 7.

Immunohistochemical localization of CT-type neuropeptide expression in *A. japonicus* using antibodies to the starfish CT-type neuropeptide ArCT.

A. Transverse section showing immunostaining in the radial nerve cord. B. Longitudinal section showing immunostaining in the radial nerve cord. C. High magnification image of a transverse section showing immunostained cells (arrowheads) and immunostained fibers in the neuropile (asterisks) of the ectoneural region of the radial nerve cord. Note the absence of immunostaining in the hyponeural region. D, E. Longitudinal sections showing immunostaining in the circumoral nerve ring and in tentacle nerves that emanate from the circumoral nerve ring. F. High magnification image of a transverse section showing immunostained cells (arrowheads) in the circumoral nerve ring. G. Longitudinal section showing immunostained fibers (asterisk) at the junction between the pharynx and the stomach. H. Longitudinal section showing immunostained fibers (arrowheads) in the intestine. I. Longitudinal section showing immunostained fibers (arrowheads) in the cloaca. Abbreviations: An, anus; BW, body wall; Cl, cloaca; CONR, circumoral nerve ring; Ec, Ectoneural region of the radial nerve cord; Hy, hyponeural region of the radial nerve cord; LM, longitudinal muscle; Lu, lumen; Mu, mucosa; RNC, radial nerve cord; TN, tentacle nerve. Scale bar, as shown in panel I, is 100 μ m in A, B, D, E, I; 50 μ m in F, G, H; 25 μ m in C.

ACh-induced contraction observed at 10^{-5} M (**Figure 8D** [↗](#)). Control experiments in which ACh-induced contraction was recorded without addition of CT-type peptides showed that ACh-induced contraction of longitudinal muscles and intestinal preparations persisted for ≥ 15 minutes, confirming that relaxation effects observed could be attributed to addition of AjCT1 or AjCT2 (**Figure 8-figure supplement 1B** [↗](#), **1C** [↗](#)).

2.6 AjCT2 affects feeding and growth in *A. japonicus*

To investigate the effects of AjCT1 and AjCT2 *in vivo*, a long-term injection experiment was carried out at concentrations of 5×10^{-3} mg/mL (low-dose group) and 5×10^{-1} mg/mL (high-dose group), with the injection volume calculated based on the wet weight of sea cucumbers at 1 μ L/g. As shown in **Figure 9A** [↗](#), there were no significant changes in weight gain rate (WGR) and specific growth rate (SGR) in the AjCT1 low concentration group (CT1L), AjCT1 high concentration group (CT1H) and AjCT2 low concentration group (CT2L) ($p \geq 0.05$), but in the AjCT2 high concentration group (CT2H) there was a significant increase in WGR and SGR in comparison with the control group (CO) ($p < 0.05$). Correspondingly, the mass of feces increased and the residual bait mass decreased (**Figure 9A-figure supplement 1** [↗](#)). Furthermore, the relative intestinal expression levels of several growth factors were investigated and the results showed that *AjGDF-8*, a growth inhibitory factor (Li et al., 2016 [↗](#)), was significantly down-regulated in all the experimental groups. In contrast, the relative expression levels of *AjIgf* and *AjMegf6*, growth promoting factors (Párrizas and LeRoith, 1997 [↗](#); Deribe et al., 2009 [↗](#)), were significantly up-regulated in intestines (**Figure 9B** [↗](#)).

Subsequently, the relative intestinal expression levels of AjCT1 and AjCT2 receptors were further tested for the CT2H group. The results showed that the relative expression levels of all three receptors were significantly up-regulated, with the ranked fold change being *AjCTR* > *AjPDRF1* > *AjPDRF2* (**Figure 9C** [↗](#)). Conversely, the relative expression levels of the three receptor genes were significantly down-regulated in *siAjCTP1/2-1* treated animals (**Figure 10A** [↗](#), **10B** [↗](#) and **10C** [↗](#)). The interference rate for *siAjCTP1/2-1* was 66 % (**Figure 10-figure supplement 1** [↗](#)).

In long-term AjCT2 loss-of-function studies, the *siAjCTP1/2-1* group exhibited significant reductions in WGR and SGR compared to the control group (*siNC*) ($p < 0.05$, **Figure 11A** [↗](#)). Correspondingly, except for phase I, the *siAjCTP1/2-1* group showed significantly increased remaining bait and decreased excrement across phases II-VI (**Figure 11B** [↗](#)). Furthermore, the growth inhibitory factor *AjGDF-8* was significantly up-regulated and the growth promoting factor *AjMegf6* was significantly down-regulated in the *siAjCTP1/2-1* group (**Figure 11C** [↗](#)).

2.7 AjPDRF2 mediates CT-type signaling-induced feeding and growth promotion in *A. japonicus*

The differential activation of signaling pathways and associated physiological outcomes between AjCT2 and AjCT1 suggested that AjCT2 may promote feeding and growth via AjPDRF2-mediated mechanisms. To investigate this hypothesis, we performed long term knockdown experiments targeting AjPDRF2. As shown in **Figure 12A** [↗](#), *siAjPDRF2-1* effectively knocked down the expression of *AjPDRF2*. In the *siAjPDRF2-1* group, both WGR and SGR were significantly decreased compared to *siNC* group (**Figure 12B** [↗](#)). Correspondingly, the remaining bait weight was significantly increased in *siAjPDRF2-1* group (except during phase I), and the weight of excrement was significantly decreased in phase V and VI (**Figure 12C** [↗](#)). Additionally, the growth promoting factor *AjIgf* was significantly down-regulated in the *siAjPDRF2-1* group (**Figure 12D** [↗](#)).

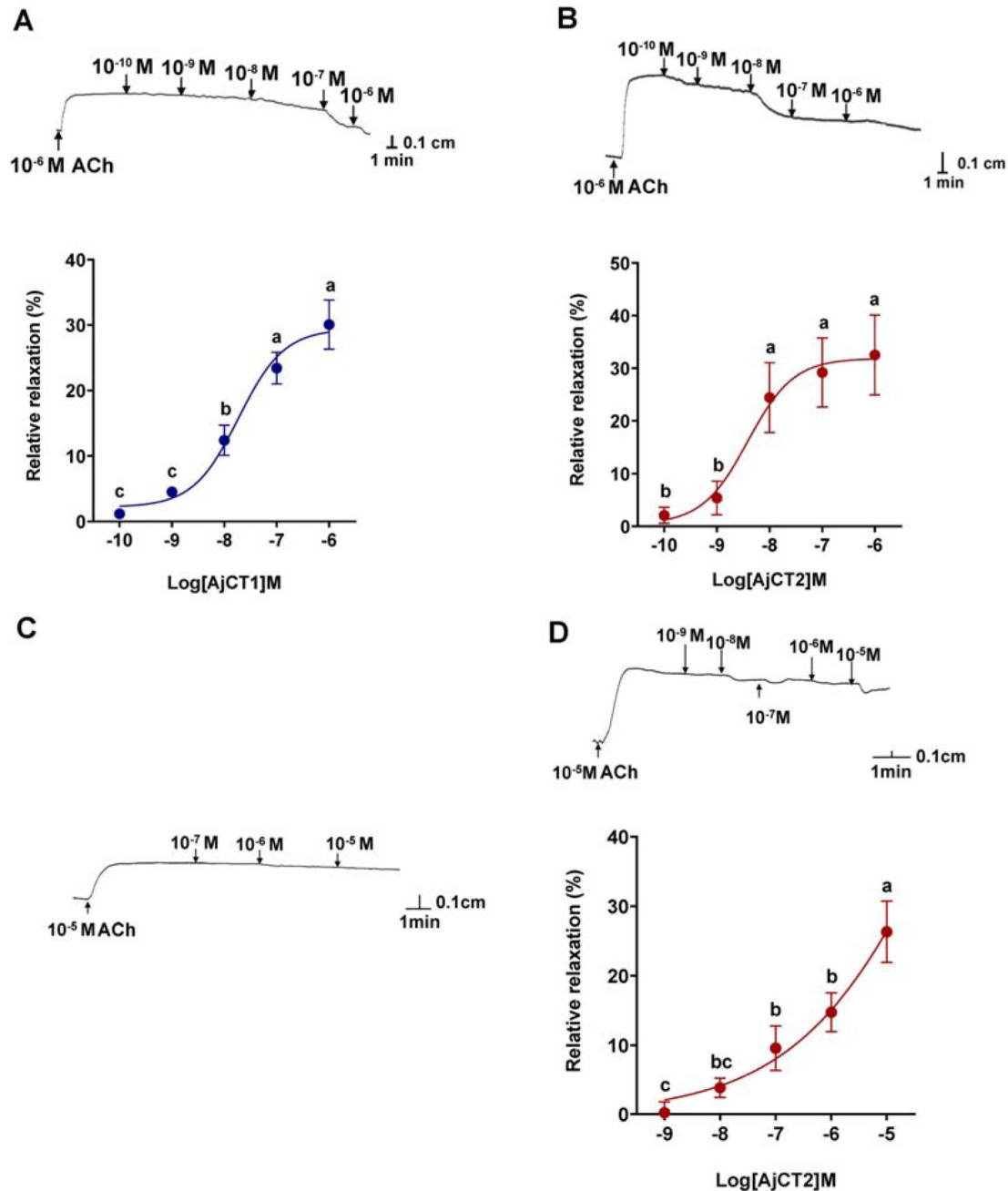


Figure 8.

Pharmacological effects of AjCT1 and AjCT2 on longitudinal muscle and intestine preparations from *A. japonicus*.

(A) and (B) show representative recordings of the relaxing effects of AjCT1 and AjCT2 on longitudinal muscle preparations. The graphs show the dose-dependent relaxing effects of AjCT1 and AjCT2 (10^{-10} – 10^{-6} M), calculated as the percentage reversal of the contracting effect of 10^{-6} M ACh. (C) and (D) show representative recordings of experiments in which AjCT1 (10^{-7} – 10^{-5} M) and AjCT2 (10^{-9} – 10^{-5} M) were tested on intestine preparations. AjCT1 had no effect, but AjCT2 caused relaxation and the graph shows the dose-dependent relaxing effect of AjCT2 (10^{-9} – 10^{-5} M), calculated as the percentage reversal of the contracting effect of 10^{-5} M ACh. All recordings shown are representative of three independent experiments. Mean values $\pm$ SEM were determined from three preparations. Lowercase letters a, b and c above columns indicate the statistical difference at $p < 0.05$, whilst there is no significant difference between bc and c.

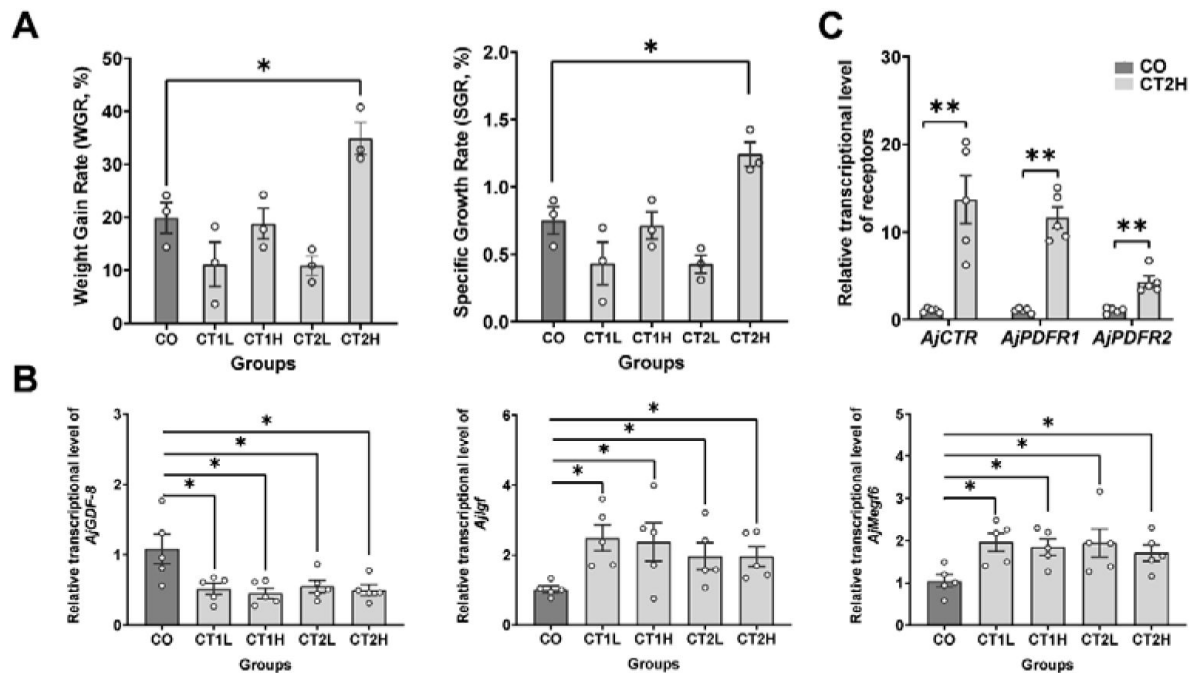


Figure 9.

***In vivo* pharmacological tests revealed that AjCT1 and AjCT2 affect feeding and growth-related gene expression in *A. japonicus*.**

(A) Weight gain rate (WGR) and specific growth rate (SGR) in experimental and control groups. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments of each group). (B) The relative intestinal transcript expression levels of *AjGDF-8*, *AjIgf* and *AjMegf6* in different groups after injection of AjCT1 or AjCT2 or sterilized seawater (CO) in *A. japonicus*. Values are means $\pm$ SEM ($n = 5$ biological duplication). (C) The relative intestinal expression levels of AjCT1/2 receptors (*AjCTR*, *AjPDRF1* and *AjPDRF2*) in CT2H and CO groups after 24 days injection experiment. Values are means $\pm$ SEM ($n = 5$ biological duplication). CO: control group (sterilized seawater); CT1L/CT2L: AjCT1/AjCT2 low concentration group (5×10^{-3} mg/mL); CT1H/CT2H: AjCT1/AjCT2 high concentration group (5×10^{-1} mg/mL). * indicates statistically significant differences with $p < 0.05$, ** indicates statistically significant differences with $p < 0.01$.

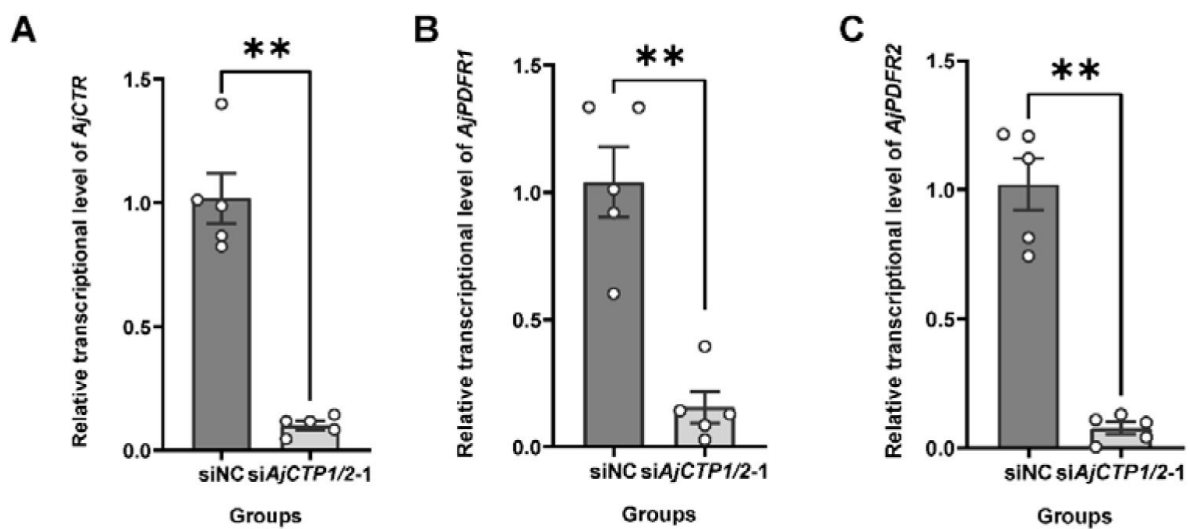


Figure 10.

The relative intestinal expression levels of *AjCT1/2* receptors after *AjCTP1/2-1* knockdown.

(A), (B) and (C) show the relative expression levels of *AjCTR*, *AjPDFR1* and *AjPDFR2* respectively. Values are means ± SEM (n = 5).

** indicates statistically significant differences with $p < 0.01$.

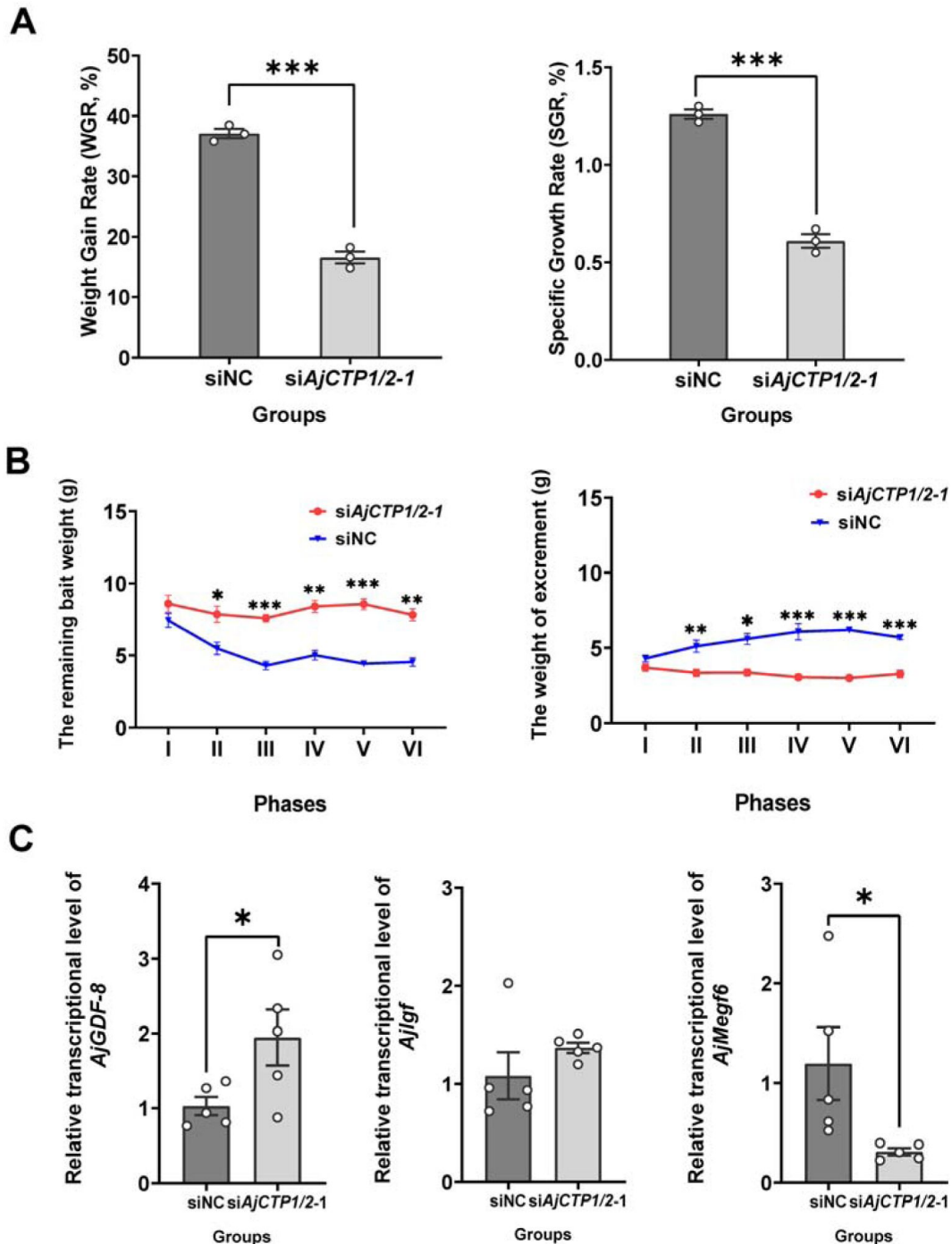


Figure 11.

Effects of 24-day *AjCTP1/2-1* knockdown *in vivo* on feeding and growth in *A. japonicus*.

(A) Weight gain rate (WGR) and specific growth rate (SGR) in siNC and siAjCTP1/2-1 groups. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments for each group). (B) Calculation of the remaining bait and excrement mass in different phases. The 24-day experiment was divided into six phases (I-VI), with each phase comprising four days. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments for each group). (C) The relative intestinal transcript expression levels of *AjGDF-8*, *AjIgf* and *AjMegf6* in siNC and siAjCTP1/2-1 groups. Values are means $\pm$ SEM ($n = 5$ biological duplication). siNC: negative control group; siAjCTP1/2-1: effectively experimental groups targeting *AjCTP1/2*. Asterisks represent statistically significant differences as follows: *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$.

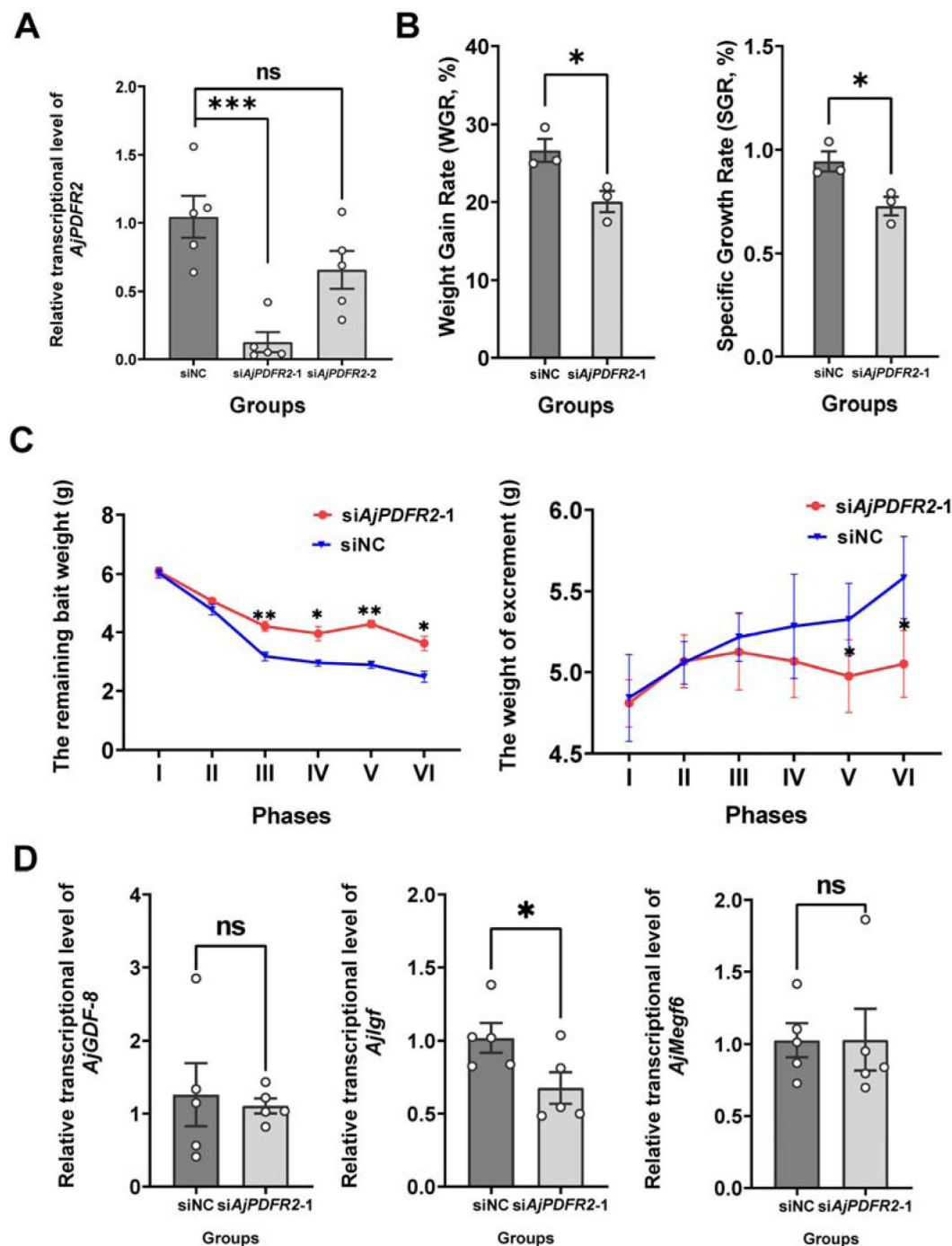


Figure 12.

Effects of *AjPDFR2*-1 knockdown *in vivo* on feeding and growth in *A. japonicus*.

(A) The relative expression level of *AjPDFR2* in siNC, si*AjPDFR2*-1 and si*AjPDFR2*-2. (B) Weight gain rate (WGR) and specific growth rate (SGR) in siNC and si*AjPDFR2*-1 groups. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments for each group). (C) The remaining bait and excrement weight in different phases. The 24-day experiment was divided into six phases (I–VI), with each phase comprising four days. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments of each group). (D) The relative intestinal transcript expression levels of *AjGDF-8*, *AjIgf* and *AjMegf6* in siNC and si*AjPDFR2*-1 groups. Values are means $\pm$ SEM ($n = 5$ biological duplication). siNC: negative control group; si*AjPDFR2*-1 and si*AjPDFR2*-2: experimental groups targeting *AjPDFR2*. Asterisks represent statistically significant differences as follows: *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$. ns: not significant.

3. Discussion

Previous studies have revealed that the CT-type neuropeptide DH₃₁ can act as a ligand for a CTR/CLR-type receptor and a pigment-dispersing factor (PDF) receptor in a protostome – the insect *D. melanogaster* (Mertens et al., 2005 [↗](#); Shafer et al., 2008 [↗](#); Schwartz et al., 2019 [↗](#)). Here we show that two CT-type peptides (AjCT1 and AjCT2) act as ligands for a CTR/CLR-type receptor and two pigment-dispersing factor (PDF) type receptors in a deuterostome – the sea cucumber *A. japonicus* (phylum Echinodermata). This is an important finding because it indicates that the ability of CT-type neuropeptides to act as ligands for both CTR/CLR-type receptors and PDF-type receptors may be an evolutionarily ancient property that originated in a common ancestor of the Bilateria. Alternatively, the ability of CT-type neuropeptides to act as ligands for PDF-type receptors in *D. melanogaster* and *A. japonicus* may have evolved independently. Further studies on a wider variety of both protostome (e.g. molluscs, annelids) and deuterostome taxa (e.g. other echinoderms, hemichordates) are needed to address this issue. Here, we also investigated the signal transduction pathways that are activated by AjCT1 and AjCT2 when these peptides bind to their receptors and then proceeded to investigate the expression and actions of these peptides to gain insights into their physiological/behavioral roles in *A. japonicus*, as discussed below.

3.1 Alternative RNA splicing gives rise to transcripts encoding one (AjCT2) or two (AjCT1, AjCT2) CT-type neuropeptides in *A. japonicus*

In vertebrates, alternative RNA splicing gives rise to transcripts encoding CT or CGRP (Amara et al., 1982 [↗](#); Morris et al., 1984 [↗](#)). Analysis of transcriptome/genome sequence data from *A. japonicus* has revealed that in this species alternative RNA splicing gives rise to transcripts encoding one (AjCT2) or two (AjCT1, AjCT2) CT-type neuropeptides (Rowe et al., 2014 [↗](#); Zandawala et al., 2017 [↗](#); Chen et al., 2019 [↗](#); this study). A similar pattern of alternative RNA splicing has been reported for CT-type precursors in other sea cucumbers (*H. scabra* and *H. glaberrima*; class Holothuroidea) and in brittle stars (*Amphiura filiformis* and *Ophiopsila Aranea*; class Ophiuroidea) (Zandawala et al., 2017 [↗](#); Suwansa-Ard et al., 2018 [↗](#)). In contrast, CT-type precursors identified in other echinoderm classes (Asteroidea, Echinoidea and Crinoidea) comprise a single CT-type neuropeptide (Rowe and Elphick, 2012 [↗](#); Mirabeau and Joly, 2013 [↗](#); Rowe et al., 2014 [↗](#); Semmens et al., 2016 [↗](#); Cai et al., 2018 [↗](#); Aleotti et al., 2022 [↗](#)). However, it remains to be determined if this feature of CT-type precursors in sea cucumbers and brittle stars evolved independently or originated in a common ancestor of eleutherozoan echinoderms but with subsequent loss in Asteroidea and Echinoidea (Figure 13 [↗](#)).

Analysis of the sequences of AjCT1 and AjCT2 revealed the presence of two cysteine residues in their N-terminal regions, which is an evolutionarily conserved feature of CT-type peptides. Structural analysis of CT-type peptides in vertebrates and in starfish has revealed that these cysteines form a disulphide bridge in the mature peptides (Cai et al., 2018 [↗](#)) and therefore it is inferred that this post-translational modification also occurs in AjCT1 and AjCT2. Furthermore, the C-terminal residue of AjCT1 and ACT2 is predicted to be an amidated proline, which is a conserved post-translational modification of CT/DH₃₁-type peptides in many taxa, but not vertebrate CGRPs (Cai et al., 2018 [↗](#)).

3.2 Identification and characterisation of receptors for CT-type neuropeptides in *A. japonicus*

Informed by previous genome-wide analysis of GPCRs in *A. japonicus* (Huang et al., 2021 [↗](#)), we identified three candidate receptors for AjCT1 and/or AjCT2. Firstly, AjCTR, which is a homolog of CT/DH₃₁-type receptors that have been identified in other taxa, including vertebrates and insects

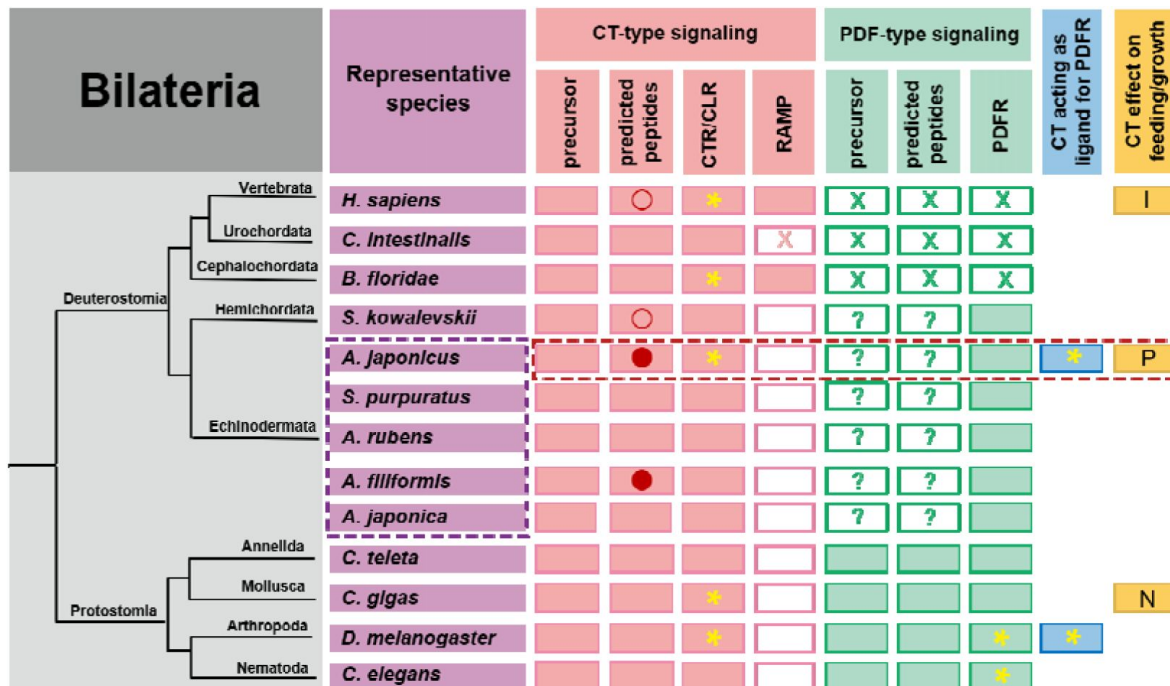


Figure 13.

Phylogenetic summary of the occurrence and properties of CT-type and PDF-type signaling systems and effects of CT-type peptides on feeding/growth regulation in representative species.

Echinoderms are boxed in purple with dashed lines and *A. japonicus*, the focus of this study, is framed in red with dashed lines. Pink filled rectangles indicate the occurrence of CT-type precursors/neuropeptides, CTR/CLR-type receptors and RAMPs. Green filled rectangles indicate the occurrence of PDF-type precursors/neuropeptides and receptors. Blue filled rectangles indicate species in which CT-type peptides have been shown to act as ligands for PDF-type receptors. Yellow filled rectangles indicate species in which effects or lack of effects of CT-type peptides on feeding/growth have been reported (I: Inhibition; P: Promotion; N: No effect). Red circles indicate that neuropeptides are generated by alternative splicing of precursor transcripts, with an unfilled circle indicating that transcripts encode one neuropeptide and with a filled circle indicating that transcripts can encode two neuropeptides in tandem. A yellow asterisk marks species in which neuropeptides have been shown to act as ligands for receptors experimentally. Unfilled rectangles indicate the absence of a protein. Unfilled rectangles containing a cross (X) indicate loss of a protein during evolution. Unfilled rectangles containing a question mark (?) indicate the existence of precursors/peptides that have yet to be identified.

(Garelja et al., 2022 [↗](#); Johnson et al., 2005 [↗](#); Toribio et al., 2003 [↗](#)). Secondly, AjPDFR1 and AjPDF2, which are homologs of the PDF-type receptor that has been shown to be activated by DH₃₁ in *D. melanogaster* (Mertens et al., 2005 [↗](#); Zandawala, 2012 [↗](#)).

In mammals, CTR and CLR mediate effects of all CT-related neuropeptides, but their ligand specificity relies on the presence and types of receptor activity-modifying proteins (RAMPs) (McLatchie et al., 1998 [↗](#); Purdue et al., 2002 [↗](#); Conner et al., 2004 [↗](#); Hay et al., 2018 [↗](#)). Thus, CTR acts as a functional receptor for CT-type peptides without an associated RAMP (Katafuchi et al., 2009 [↗](#)), whereas activation of downstream signaling via CLR requires an associated RAMP (Hay et al., 2018 [↗](#)). Here, we investigated the occurrence of RAMPs in *A. japonicus* using the tblastn method employed by Sekiguchi et al., (2016 [↗](#)), which enabled identification of RAMPs in the cephalochordate *B. floridae*. However, no RAMPs were identified in *A. japonicus*, suggesting that AjCTR may be functionally more similar to CTR than CLR. Accordingly, tblastn analysis of other available echinoderm genomes (*A. rubens*, *S. purpuratus* and *Anneissia japonica*), as well as transcriptome databases for brittle stars (*O. victoriae*, *O. aranea* and *A. filiformis*), also failed to identify RAMPs. Likewise, RAMPs have also not been found in protostomes (Schwartz et al., 2019 [↗](#)). Therefore, RAMPs may be unique to chordates, although interestingly RAMPs have also not been identified in the urochordate *C. intestinalis* (Sekiguchi et al., 2009 [↗](#)) (**Figure 13** [↗](#)).

As highlighted above, here we show that the CT-type peptides AjCT1 and AjCT2 both act as ligands for a CTR/CLR-type receptor (AjCTR) and two PDF-type receptors (AjPDFR1, AjPDFR2) in *A. japonicus*. Furthermore, characterization of downstream signaling mechanisms revealed that both AjCT1 and AjCT2 trigger AjCTR-mediated cAMP accumulation, Ca²⁺ mobilization, and ERK signaling. These findings are consistent with previous studies on CT-type receptors in vertebrates and in the mollusc *Magallana gigas* (Pondel, 2000 [↗](#); Walker et al., 2010 [↗](#); Andreassen et al., 2014 [↗](#); Schwartz et al., 2019 [↗](#)). Both AjCT1 and AjCT2 trigger cAMP accumulation, Ca²⁺ mobilization, and ERK signaling mediated by AjPDFR1. Both AjCT1 and AjCT2 triggered cAMP accumulation mediated by AjPDFR2, but only AjCT2 (not AjCT1) triggered Ca²⁺ mobilization and ERK signaling mediated by AjPDFR2. These differences in the signal transduction mechanisms activated by AjCT1 and AjCT2 may account for differences in the pharmacological activities and physiological roles of these peptides, as discussed below.

3.3 AjCT1 and AjCT2 act as muscle relaxants in *A. japonicus*

Analysis of the expression of the gene encoding the precursors comprising AjCT2 or AjCT1 and AjCT2 revealed the occurrence of transcripts in the circumoral nervous system as well as in peripheral tissues/organs, including the longitudinal muscle, digestive system and gonads. Immunohistochemical analysis using an antiserum to the starfish (*A. rubens*) CT-type neuropeptide ArCT enabled visualisation of CT-type neuropeptide(s) in juvenile specimens of *A. japonicus*. Intense and extensive immunostaining was revealed in the ectoneural region of the radial nerve cords and circumoral nerve ring, consistent with the high levels *AjCTP1/2* transcripts detected in circumoral nervous tissue. Immunostained nerve fibers were also observed in the digestive system (including the intestine), consistent with the detection *AjCTP1/2* transcripts in intestine. However, no immunostaining was observed in longitudinal muscles, which is inconsistent with the detection of *AjCTP1/2* transcripts this tissue. This may reflect differences in the detection sensitivity of the techniques used. In accordance with the expression pattern of *AjCTP1/2* in *A. japonicus*, analysis of CT-type neuropeptide expression in other echinoderms has revealed a widespread pattern of expression in larval and adult starfish (*A. rubens*) and in juvenile and adult featherstars (*Antedon mediterranea*) (Mayorova et al., 2016 [↗](#); Cai et al., 2018 [↗](#); Aleotti et al., 2022 [↗](#)).

Several CT-related peptides act as muscle relaxants in vertebrates (Steiner et al., 1991 [↗](#); Pinto et al., 1996 [↗](#); Yoshimoto et al., 1998 [↗](#); Edvinsson et al., 2001 [↗](#); Golpon et al., 2001 [↗](#); Kandilci et al., 2008 [↗](#)) and, accordingly, previous studies have revealed that CT-type neuropeptides cause relaxation of muscle preparations in starfish, including locomotory organs (tube feet) and the

apical muscle, which is functionally similar to the longitudinal muscle in sea cucumbers (Cai et al., 2018). Therefore, it was of interest to investigate if AjCT1 and/or AjCT2 act as muscle relaxants in *A. japonicus*. Both AjCT1 and AjCT2 caused dose-dependent relaxation of longitudinal muscle preparations, consistent with the relaxing effect of CT-type peptides on starfish apical muscle preparations. It is worth noting that while AjCT2 at 10^{-6} M exhibited the maximum efficacy in reversing ACh-induced contraction, its relaxing effect weakened compared to the 10LL M treatment when applied at progressively higher concentrations. This diminished efficacy likely stems from receptor desensitization, induced by repeated AjCT2 application to longitudinal muscle preparations (Tsurumaki et al., 2003; Arrowsmith and Wray, 2014). Furthermore, AjCT2, but not AjCT1, caused dose-dependent relaxation of intestine preparations from *A. japonicus*. Interestingly, the CT-type peptide ArCT does not to cause relaxation of cardiac stomach preparations from the starfish *A. rubens* (Cai et al., 2018). Therefore, from a functional perspective ArCT is more like AjCT1 than AjCT2 and we speculate that this may reflect loss of an exon encoding a AjCT2-like peptide in the Asteroidea lineage. Further insights into this issue and the evolution of CT-type neuropeptide function echinoderms could be obtained by investigation of the actions of CT-type peptides in brittle stars and sea urchins. Differences in the effects of AjCT1 and AjCT2 on the intestine may be explained by differences in their signal transduction mechanisms. Thus, only AjCT2, not AjCT1, triggered AjPDR2-mediated activation of Ca^{2+} mobilization and ERK signaling and therefore we speculate that the relaxing effect of AjCT2 on the intestine may be mediated by activation of AjPDR2.

3.4 AjCT2/AjPDR2 stimulates feeding and growth in *A. japonicus*

In vivo pharmacological assays demonstrated that only high concentrations of AjCT2 significantly enhanced feeding and growth rates in *A. japonicus*. In contrast, neither a low concentration of AjCT2 nor any concentration of AjCT1 (low or high) induced detectable effects. Furthermore, long-term knockdown of *AjCTP1/2* further validated the essential role of AjCT2 in regulating feeding and growth in this species. To elucidate the receptor mediating AjCT2's feeding-and growth-promoting effects, we selected AjPDR2 based on its distinct activation profile: AjCT2 selectively activated AjPDR2, inducing downstream ERK1/2 phosphorylation, whereas AjCT1 exhibited no activity toward this receptor. Given this receptor specificity, we performed *AjPDR2* knockdown experiments, which revealed phenotypic changes consistent with those in *AjCTP1/2* knockdown animals, including significantly reduced WGR and SGR, alongside increased remaining bait accumulation and diminished excrement output compared to control. Collectively, these results support a model wherein AjCT2 promotes feeding and growth in *A. japonicus* via AjPDR2-dependent activation of the cAMP/PKA/ERK1/2 and Gαq/Ca²⁺/PKC/ERK1/2 cascades. Considering the inherent complexity of neuropeptide signaling systems, which involve multiple GPCR subtypes coupled to diverse signaling cascades, ligands bound to the same receptor may activate distinct G protein subforms within a single cell (Møller et al., 2003; Mendel et al., 2020). Receptor activation modes may be modulated by structural polymorphisms or binding site diversity (Wong et al., 2000; Changeux, 2010), as well as by the differential efficacy of peptides in activating receptors *in vivo*.

Collectively, these findings indicate that AjCT2 has a physiological role in stimulating feeding and growth in *A. japonicus*. To the best of our knowledge, this is the first study to report orexigenic and growth-promoting effects of a CT-type neuropeptide in bilaterian animals. Interestingly, previous studies have reported the expression of CT-type neuropeptides in the digestive system of invertebrate chordates (Sekiguchi et al., 2009; Sekiguchi et al., 2016). However, in contrast to the stimulatory effect of AjCT2 on feeding/growth in *A. japonicus*, CT-type neuropeptides have been found to inhibit feeding and growth in vertebrates (Del et al., 2002; Hay et al., 2018; Nakamura et al., 2018). Thus, it appears that opposing roles of CT-type neuropeptide signaling systems as regulators for feeding/growth have evolved in deuterostomes, with a stimulatory role reported here in an echinoderm and an inhibitory role reported previously in vertebrates (Figure 13). Further research is now needed to investigate if the stimulatory effect of AjCT2 on

feeding/growth observed here in *A. japonicus* also applies to other sea cucumber species. Moreover, our discovery of this effect of AjCT2 on *A. japonicus* may provide a basis for development of novel approaches to improve aquaculture of this economically important species.

4. Materials and methods

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Cell line (<i>Homo sapiens</i>)	HEK293 cell line	Procell	CL-0005	
Antibody	phosphorylated ERK1/2 kinases (Thr202/Tyr204) (goat)	Cell Signaling Technology	CAT#4930	IF (1:2000)
Antibody	ERK1/2 kinases (137F5) (goat)	Cell Signaling Technology	CAT#4695T	IF (1:1000)
Antibody	HRP-conjugated goat anti-rabbit IgG (goat)	Absin	Abs20040ss	IF (1:2000)
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction-- AjCTR	This paper	PCR PRIMER	FORWARD CGCGGATCCAAAATGCAGGAGA ACGATTC BamH ☐
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction-- AjCTR	This paper	PCR PRIMER	REVERSE CCGGAATTCCTAAACTACCGTT GTTTTAT EcoR ☐
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction— AjPDFR1	This paper	PCR PRIMER	FORWARD CGCGGATCCACAATGAGTGTCC GGAACGT BamH ☐
sequence-based reagent	Primers for pcDNA3.1(+) plasmid	This paper	PCR PRIMER	REVERSE CCGGAATTCATTTCCCACCCC ACTCAC

	construction— AjPDFR1			EcoR □
sequence- based reagent	Primers for pcDNA3.1(+) plasmid construction— AjPDFR2	This paper	PCR PRIMER	FORWARD CGCGGATCCAAGATGGCGAACA ATGAAAC BamH □
sequence- based reagent	Primers for pcDNA3.1(+) plasmid construction— AjPDFR2	This paper	PCR PRIMER	REVERSE CCGGAATTCTCATTCAAATT TGCCGT EcoR □
sequence- based reagent	Plasmid construction AjCTR-EGFP	This paper	PCR PRIMER	FORWARD CCGGAATTCTGAAAATGCAGGA GAACGATTC EcoR □
sequence- based reagent	Plasmid construction AjCTR-EGFP	This paper	PCR PRIMER	REVERSE CGCGGATCCCGAACTACCGTTG TTTATCTG BamH □
sequence- based reagent	Plasmid construction AjPDFR1-EGFP			CCGGAATTCTGACAATGAGTGT CGGGAACGT EcoR □
sequence- based reagent	Plasmid construction AjPDFR1-EGFP			CGCGGATCCCGTTTCCACCCC ACTACTGT BamH □
sequence- based reagent	Plasmid construction AjPDFR2-EGFP			CCGGAATTCTGAAGATGGCGAA CAATGAAAC EcoR □
sequence- based reagent	Plasmid construction AjPDFR2-EGFP			CGCGGATCCCGTTCATTAAATTT GCCGTTCA BamH □
sequence- based reagent	Primers for qPCR assays β -Actin	This paper	PCR PRIMER	FORWARD AAGGTTATGCTCTTCCTCAGCT
sequence- based reagent	Primers for qPCR assays β -Actin	This paper	PCR PRIMER	REVERSE GATGTCACGGACGATTCACG

sequence-based reagent	Primers for qPCR assays <i>β-Tubulin</i>	This paper	PCR PRIMER	FORWARD GAAAGCCTTACGACGGAACA
sequence-based reagent	Primers for qPCR assays <i>β-Tubulin</i>	This paper	PCR PRIMER	REVERSE CACCACGTGGACTCAAATG
sequence-based reagent	Primers for qPCR assays <i>AjCTP1/2</i>	This paper	PCR PRIMER	FORWARD GTTTGGCGGTACCCATTAC
sequence-based reagent	Primers for qPCR assays <i>AjCTP1/2</i>	This paper	PCR PRIMER	REVERSE GGCGTTCGGTCTGTTAATAG
sequence-based reagent	Primers for qPCR assays <i>AjMegf6</i>	This paper	PCR PRIMER	FORWARD CGGTCCTTACTGTGCGGAGA
sequence-based reagent	Primers for qPCR assays <i>AjMegf6</i>	This paper	PCR PRIMER	REVERSE CGTCCTCCCAGCTGGACATC
sequence-based reagent	Primers for qPCR assays <i>AjIgf</i>	This paper	PCR PRIMER	FORWARD TGGAAGGCAGTAAGGCTA
sequence-based reagent	Primers for qPCR assays <i>AjIgf</i>	This paper	PCR PRIMER	REVERSE TAGGCTGATGTGGTTGGA
sequence-based reagent	Primers for qPCR assays <i>AjGDF-8</i>	This paper	PCR PRIMER	FORWARD ATACGCCCATACTCACATCC
sequence-based reagent	Primers for qPCR assays <i>AjGDF-8</i>	This paper	PCR PRIMER	REVERSE TGGCAGGAGTCAACCTTCAT
sequence-based reagent	Primers for qPCR assays <i>AjCTR</i>	This paper	PCR PRIMER	FORWARD GACGAATGGAGGAGAAGG
sequence-based reagent	Primers for qPCR assays <i>AjCTR</i>	This paper	PCR PRIMER	REVERSE CGCCGCAATTAGAGTAGT

sequence-based reagent	Primers for qPCR assays <i>AjPDFR1</i>	This paper	PCR PRIMER	FORWARD TTATCAACCGCCGTCTTC
sequence-based reagent	Primers for qPCR assays <i>AjPDFR1</i>	This paper	PCR PRIMER	REVERSE CAGGAGGAGGAACACATTC
sequence-based reagent	Primers for qPCR assays <i>AjPDFR2</i>	This paper	PCR PRIMER	FORWARD TGTCTCTCTGCCTCTTCT
sequence-based reagent	Primers for qPCR assays <i>AjPDFR2</i>	This paper	PCR PRIMER	REVERSE GTCCTTCTCATCGGTGTG
sequence-based reagent	Primers for 3'RACE <i>AjCTP1</i> <i>AjCTP2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTCGAGCTG TGTCGATAACATTACGCATGG
sequence-based reagent	Primers for 3'RACE <i>AjPDFR1</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGAGGGAA AGCCAAACCAGCGAGACG
sequence-based reagent	Primers for 3'RACE <i>AjPDFR2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGTGCAAC CGGACGAAGATCCACGTG
sequence-based reagent	Primers for 5'RACE <i>AjPDFR1</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTCGTCTCG CTGGTTTGGCTTTCCCTC
sequence-based reagent	Primers for 5'RACE <i>AjPDFR2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGCCTTCC ACAAACATCCATGCGAACGC
sequence-based reagent	Primers for 5'RACE <i>AjCTR</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTACGCCGG GCTCTCAAGAAGTCATTC
sequence	RNAi	This paper	si <i>AjCTP1</i> -1	GGAAUUAUCAAGACAGUAU (dT)(dT) AUACUGUCUUGAUAAUCC (dT)(dT)
sequence	RNAi	This paper	si <i>AjCTP1</i> -2	GAAAGUCGCUAUGCAGUA (dT)(dT) UACUGCAUUAGCGACUUUC

				(dT)(dT)
sequence	RNAi	This paper	si <i>AjPDR2</i> -1	GUAGCGAUUGGUCAAUAA (dT)(dT) UUAUUCGACCAAUCGCUAC (dT)(dT)
sequence	RNAi	This paper	si <i>AjPDR2</i> -2	CAGUCAUUGUUAUCUACCA (dT)(dT) UGGUAGAUAAACAUGACUG (dT)(dT)
sequence	RNAi	This paper	siNC	UUCUCCGAACGUGUCACGU (dT)(dT) ACGUGACACGUUCGGAGAA (dT)(dT)
commercial assay or kit	SMARTer RACE 5'/3' Kit	Takara	Cat No. 634858	
commercial assay or kit	FastPure Gel DNA Extraction Mini Kit	Vazyme	DC301-01	
commercial assay or kit	Hifair® □ 1st Strand cDNA Synthesis SuperMix	YEASEN	Cat No. 11141ES60	
commercial assay or kit	Hieff UNICON Universal Blue qPCR SYBR Green Master Mix	YEASEN	Cat No. 11184ES08	
commercial assay or kit	Firefly luciferase reporter gene assay kit	MK	MF4001	
commercial assay or kit	cAMP Assay kit	R&D systems	Cat No. KGE002B	

4.1 Animals

Adult sea cucumbers were collected from an aquaculture breeding farm in Weihai, Shandong, China. All the animals were acclimated in breeding aquaria (18 L, 32 ppt) for at least seven days and fed daily with a commercially formulated diet. Tissues used for cDNA cloning and qRT-PCR were obtained from six adult *A. japonicus* (~120 g) and kept at -80 L until use. Juvenile specimens of *A. japonicus* (~15 mm in length) were used for immunohistochemical analysis of CT-type neuropeptide expression. *In vitro* functional analysis and *in vivo* experiments were carried out using adult sea cucumbers with body masses of ~120 g and ~20 g, respectively.

Ethical approval

All methods were carried out in accordance with the ARRIVE guidelines (<https://arriveguidelines.org>) for animal experiments. All animal care and use procedures were approved by the Institutional Animal Care and Use Committee of Ocean University of China (Permit Number: 20141201). They were performed according to the Chinese Guidelines for the Care and Use of Laboratory Animals (GB/T 35892-2018).

4.2 Molecular structure and phylogenetic analysis of CT-type neuropeptides and precursors

A. japonicus CT-type neuropeptide precursor sequences (*AjCTP1*, GenBank accession number: MF401985; *AjCTP2*, GenBank accession number: MF401986) were identified in our previous study (Chen et al., 2019). Based on the published sequences, transcript-specific primers for *AjCTP1* and *AjCTP2* (Key resources table) were designed using Primer3Plus (<https://www.primer3plus.com/>) and sequence accuracy was confirmed by SMARTer RACE 5'/3' Kit (TaKaRa, Kusatsu, Japan) according to the manufacturer's instruction. A sequence translation tool (<https://web.expasy.org/translate/>) was used to obtain the corresponding amino acid sequences encoded by the cloned cDNAs. To identify the conserved features of CT-type neuropeptides in Bilateria, sequences were aligned using Clustal-X with default settings and then predicted using the online tool Multiple Sequence Comparison Display Tool (<http://www.bio-soft.net/sms/index.html>). To investigate relationships of *AjCTP1*, *AjCTP2* and CT-type precursors from other Bilateria, phylogenetic analyses were performed using the Neighbor joining method. The precursor sequences were aligned using the MUSCLE plugin (iterative, 10 iterations, UPGMB as clustering method), a phylogenetic tree was generated using the neighbor joining method with MEGA7 (v.7.170509) and bootstrap values were calculated from 1000 replicate analysis and then modified using iTOL (<https://itol.embl.de/login.cgi>). The accession numbers or citations for all sequences used for analysis are listed in **Supplementary table 1**.

4.3 Cloning, sequence analysis and phylogenetic tree construction for candidate *AjCT1/2* receptors in *A. japonicus*

Five potential receptors for *AjCT1* and *AjCT2* in *A. japonicus* (GenBank accession number: PIK49998.1, PIK49999.1, PIK49604.1, PIK34007.1 and PIK62458.1) were identified based on genome-wide analysis (Huang et al., 2021). Sequence comparison revealed that both PIK49998.1 and PIK62458.1 were partial sequences of PIK34007.1. Thus, three candidate receptors belonging to Class B (Secretin-like) GPCRs named as *AjCTR* (Calcitonin receptor, GenBank accession number: PIK49604.1), *AjPDFR1* (Pigment-dispersing factor receptor 1, GenBank accession number: PIK34007.1), and *AjPDFR2* (Pigment-dispersing factor receptor 2, GenBank accession number: PIK49999.1) were predicted and then analyzed. To confirm the accuracy of these sequences, cDNAs were cloned using the SMARTer RACE 5'/3' Kit (Takara, Cat No. 634858) and then sequenced (Beijing Genomics Institute, China).

Phylogenetic relationships of *AjCTR*, *AjPDFR1* and *AjPDFR2* with GPCRs belonging to the CT-type (including DH31 and Cluster A) and PDF-type receptor families in other bilaterian animals were investigated, with CRH-type receptors used as an outgroup. Receptor sequences (see **supplementary table 2**) were aligned using MUSCLE 5.1 with the PPP algorithm (Edgar, 2022). The alignment was trimmed with the gappyout option using trimAl 1.2rev59 (Capella-Gutiérrez et al., 2009), and maximum-likelihood phylogenetic trees were generated with IQ-tree 2.0.7 for Linux 64-bit using the SH-aLRT and ultrafast bootstrap (UFBoot) methods with nearest neighbor interchange (NNI) correction tests (Guindon and Gascuel, 2003; Guindon et al., 2010; Hoang et al., 2018; Minh et al., 2020; Nguyen et al., 2015). The VT+F+I+G4 amino acid

substitution model was selected, with branch support of 1000 bootstrap replicates. The tree was rooted using CRH-type receptors as the outgroup, and edited using Treeviewer version 2.2.0 (Bianchini and Sanchez-Baracaldo, 2024 [DOI](#)).

In addition, the online tool NetPhos-3.1 (<https://services.healthtech.dtu.dk/service.php?NetPhos-3.1> [DOI](#)) was used to predict potential phosphorylation sites. The conserved domains of protein sequences were predicted by the Batch SMART plugin of TBtools (Toolbox for biologists, v.1.098652) software. For analysis of gene synteny, neighboring genes of *AjCTR*, *AjPDRF1* and *AjPDRF2* were found based on an *A. japonicus* chromosome-level genome assembly (Wang et al., 2022 [DOI](#)) and the corresponding genes in the starfish *Asterias rubens* and the sea urchin *Lytechinus variegatus* were also identified by performing BLASTp in the NCBI database. Specific primers for cloning of cDNAs encoding the candidate receptors are shown in the Key resources table and the details of the sequences used for constructing the phylogenetic tree are listed in **Supplementary table 2** [DOI](#).

4.4 Testing AjCT1 and AjCT2 as ligands for candidate receptors and exploration of downstream signaling pathways

AjCT1 and AjCT2 (both with a disulfide bridge between the two cysteine residues and with C-terminal amidation) were custom synthesized by GL Biochem Ltd. (Shanghai, China) with >95% purity. To test candidate receptors for CT-type neuropeptides and investigate their downstream activation mechanisms the following assays were used to measure cAMP signaling or Ca²⁺L signaling: i). direct measurement of cAMP levels using ELISA ii). indirect measurement of cAMP signaling using a cAMP response element-Luciferase (CRE-Luc) assay that measures cAMP-induced gene expression, iii). serum response element (SRE)-Luc assays that measure Ca²⁺-induced gene expression, iv). receptor translocation assays and v).

Western blotting

AjCTR, *AjPDRF1* and *AjPDRF2* incorporating enhanced green fluorescent protein (EGFP) were constructed respectively according to the method described by Li et al. (2022) [DOI](#). Briefly, the complete ORFs of candidate receptors were cloned and sequenced. Then, the PCR products were digested with EcoRI and BamHI (TaKaRa, Kusatsu, Japan) and then incorporated into the pEGFP-N1 vector (YouBio, Changsha, China) that was subject to the same double-digestion. Finally, the constructed plasmid with correct sequences was transfected into HEK293T cells seeded on glass coverslips in 12-well plates. After incubation with AjCT1 (10⁻⁶ M), AjCT2 (10⁻⁶ M) or serum-free DMEM (Biosharp, Hefei, China) for 15 min, cells were washed with PBS (Biosharp, Hefei, China) and fixed with 4% PFA for 10 min. Finally, cell nuclei were stained with DAPI (G-CLONE, Beijing, China) for 5 min and an Olympus bright field light microscope was used for image capture. The pcDNA 3.1(+) vectors (YouBio, Changsha, China) comprising the complete ORF of the three candidate receptors were also constructed as described by Li et al. (2022) [DOI](#). Then, they were co-transfected with the cAMP reporter pCRE-Luc (YouBio, Changsha, China) or Ca²⁺ signaling reporter pSRE-Luc plasmid including a PKC reporter gene (YouBio, Changsha, China) into HEK293T cells. Incubation was carried out with AjCT1 (10⁻⁶ M), AjCT2 (10⁻⁶ M) or serum-free DMEM (CO, negative control) for 4 h at 37 °C. Finally, luciferase activity was tested using the firefly luciferase assay kit (MK, Shanghai, China). When required, transfected cells described above were co-incubated with H89 (PKA inhibitor) (Absin, Shanghai, China), FR900359 (Gaq protein inhibitor) (Cayman Chemical, Ann Arbor, Michigan) or Gö 6983 (PKC inhibitor) (Cayman Chemical, Ann Arbor, Michigan) before the experiment to explore the potential signaling pathways. To confirm the concentration of intracellular cAMP after stimulation, HEK293T cells expressing pcDNA 3.1(+)/*AjCTR*, pcDNA 3.1(+)/*AjPDRF2* or pcDNA 3.1(+)/*AjPDRF1* were treated with AjCT1 or AjCT2 (10⁻⁹ – 10⁻⁵ M), respectively, for 15 min after 2 h starvation in serum-free medium. Subsequently, cAMP was measured using an ELISA-based cAMP assay kit (R&D systems, USA&Canada) following the

manufacturer's instruction. HEK293T cells transfected with empty pcDNA3.1(+) were used as a negative control. The specific primers used for constructing plasmids are listed in Key resources table.

Western blotting was performed as described previously (Li et al., 2022 [↗](#)). Briefly, HEK293T cells transfected with the constructed pcDNA 3.1(+) plasmid were starved in serum-free medium for 2 h, and then incubated with AjCT1 or AjCT2 (10^{-6} M). PVDF membranes containing proteins were blocked with 5% BSA (G-CLONE, Beijing, China) and incubated overnight with antibodies to phosphorylated ERK1/2 kinases (CST, Boston, China). After that, goat anti-rabbit horseradish peroxidase-conjugated secondary antibody (Absin, Shanghai, China) was applied and incubated, followed by the addition of Omni-ECL™ Femto Light Chemiluminescence reagents (EpiZyme, Shanghai, China). Finally, RVL-100-G (ECHO, America) was applied for visualization of the PVDF membrane. When required, transfected cells were co-incubated with PKA or PKC inhibitor before the experiment to identify signaling pathways that activate downstream ERK1/2 cascade.

4.5 Total RNA extraction and Quantitative

Real-Time PCR (qRT-PCR) detection

qRT-PCR was employed to detect *AjCTR*, *AjPDFR1*, *AjPDFR2*, *AjCTP1* and *AjCTP2* transcript expression in five tissues from *A. japonicus*, including the intestine, longitudinal muscle, respiratory tree, gonad and circumoral nervous system, and the circumoral nervous system was selected as the control group. Because it was difficult to design specific primers for *AjCTP1* and *AjCTP2* (Zheng et al., 2022 [↗](#)), total transcript expression levels for *AjCTP1/2* (*AjCTP1* + *AjCTP2*) were investigated. Total RNA was extracted from the collected tissues using Trizol RNA isolation reagent (Vazyme, Nanjing, China) and Hifair® L 1st Strand cDNA Synthesis SuperMix (YEASEN, Shanghai, China) was used to remove genomic DNA and synthesize cDNA according to manufacturer's instructions. For qRT-PCR, amplification was performed in 20 µL volume reactions based on Hieff UNICON Universal Blue qPCR SYBR Green Master Mix (YEASEN, Shanghai, China) and expression levels were assessed by a Corbett Rotor-Gene Q (Qiagen, Germany). *β*-actin (GenBank accession number: PIK61412.1) and *β*-tubulin (GenBank accession number: PIK51093) were used as house-keeping genes for normalization (Zhao et al., 2014 [↗](#)). The $2^{-\Delta\Delta CT}$ method was applied to analyze the relative expression levels. Specific sequences of primers used for qRT-PCR are listed in Key resources table.

4.6 Localization of CT-type neuropeptide expression

in *A. japonicus* using immunohistochemistry

Juvenile specimens of *A. japonicus* (~15 mm in length) were fixed for 24 hours at room temperature or for 48 hours at 4 °C in Bouin's fixative (75 mL saturated picric acid in seawater, 25 mL 37% formaldehyde, 5 mL acetic acid) days and then washed in water before being transferred to 50% ethanol followed by 70% ethanol. The fixed specimens were wrapped in tissue paper soaked in 70% ethanol and then transported from China to the UK. Following dehydration through 90% and 100% ethanol and clearing in xylene, specimens were embedded in paraffin wax and then sectioned (9 or 10 µm) transversely or longitudinally using a Leica RM2145 microtome. Sections were mounted on chrome alum-gelatin coated glass slides and then after incubation in xylene to remove wax followed by 100% ethanol, slides were incubated in 1% hydrogen peroxide in phosphate buffered saline (PBS, pH 7.3) to quench endogenous peroxidases. After using an ethanol series to rehydrate sections, slides were blocked in 5% goat serum in PBS containing 0.1% Tween-20 (PBST). Then slides were incubated overnight at 4 °C with the ArCT antiserum diluted to 1:2000, 1:3000 or 1:4000 in PBST. Following washing in PBST, slides were incubated with Peroxidase-AffiniPure Goat Anti-Rabbit IgG (H + L) conjugated to Horseradish Peroxidase (RRID:AB_2313567 [↗](#); Jackson ImmunoResearch, West Grove, PA) diluted 1:1000 in 2% goat serum/PBST for 3 h. Bound antibodies were revealed using a solution of 0.05% diaminobenzidine (VWR Chemicals, UK), 0.05% nickel chloride (Sigma-Aldrich, Gillingham, UK), 0.015% hydrogen

peroxide (VWR Chemicals, UK) diluted in PBS. When staining was observed, slides were washed in water and dehydrated through an ethanol series into xylene and mounted with coverslips over DPX mounting medium (Thermo Fisher Scientific). The specificity of immunostaining was assessed by pre-absorption of the ArCT antiserum (at 1:100 dilution) for 1 h at room temperature with the ArCT antigen peptide (25 μ M) prior to testing on sections at 1:3000 dilution. Images of stained sections were taken using a QIClick CCD Colour Camera (Qimaging, UK) linked to a DMRAI light microscope (Leica Microsystems, UK) with Volocity v. 6.3.1 image analysis software (Perkin-Elmer, USA) running on an iMac computer (27 inch with OS Yosemite, v. 10.10). Montages of images were prepared and labelled using GIMP (version 3.0.2) and Adobe Photoshop (version 26.7) running on a MacBook Pro laptop (OS Sequoia 15.5).

4.7 Analysis of the in vitro activity of AjCT1 and AjCT2 on preparations from *A. japonicus*

Longitudinal muscles and intestines (20 mm in length) from *A. japonicus* were dissected and then suspended in a 50 mL glass organ bath containing 20 mL sterilized seawater. Both ends of the preparations were tied with cotton ligatures and then one end was tied to a metal hook and the other end was connected with a High Grade Isotonic Transducer (ADInstruments MLT0015, Oxford, UK). Before testing, the preparation was stabilized for about 20 min, and the resting tension was adjusted to 1.0 g. Informed by previous findings from Cai et al. (2018) [\[14\]](#), we investigated if AjCT1 and AjCT2 caused relaxation of preparations. First, acetylcholine (ACh) was applied to induce the contraction of preparations and then when a stable state of contraction was achieved, 20 μ L AjCT1 or AjCT2 at a range of concentrations was added and relaxation responses were recorded. Output signal (cm) from PowerLab data acquisition hardware (ADInstruments PowerLab 4/26, Oxford, UK) was calculated using LabChart (v8.0.7) software.

4.8 Investigation of the in vivo pharmacological effects of AjCT1 and AjCT2 in *A. japonicus*

A total of 150 adult sea cucumbers were randomly assigned to five groups (30 individuals per group), including two high concentration groups (CT1H and CT2H, 5×10^{-1} mg/mL), two low concentration groups (CT1L and CT2L, 5×10^{-3} mg/mL) and the control group (CO, sterilized seawater), with each group consisting of three parallel experiments comprising ten animals. Before the experiments, sea cucumbers were starved for a day to normalize their physiological status and their wet weights were measured. AjCT1 and AjCT2 were dissolved in sterilized seawater and then injected through the peristome at noon every other day (1 μ L/g wet weight) and the CO group were injected in the same way but with sterilized seawater (1 μ L/g wet weight). Feces and residual bait were collected and weighed after drying every day. After 24 days, wet weights of sea cucumbers were measured again. Then, WGR and SGR were calculated to assess their feeding and growth conditions. Formulas are as follows: $WGR = 100 \times (W_t - W_0) / W_0$, $SGR = 100 \times (\ln W_t - \ln W_0) / D$. In the formula, W_0 represents the initial wet body weight (g), W_t represents the final wet body weight (g), D represents the feeding days. Subsequently, animals were dissected and intestines were collected for detecting the expression levels of the three receptor genes (*AjCTR*, *AjPDFR1*, *AjPDFR2*) and several growth factors, including multiple epidermal growth factor 6 (*AjMegf6*, GenBank accession number: MG018199.1), insulin-like growth factor (*AjIgf*, GenBank accession number: PIK50518.1) and growth and differentiation factor-8 (*AjGDF-8*, GenBank accession number: KP100064.1) (Li et al., 2016 [\[15\]](#); Wang et al., 2018 [\[16\]](#)). The specific primers for qRT-PCR analysis are listed in the Key resources table.

4.9 Loss-of-function experiments by RNA interference (RNAi)

To investigate how receptor expression is affected by knock-down of AjCTP expression, an RNAi experiment was carried out. It was difficult to design *AjCTP1*- and *AjCTP2*- specific interference oligonucleotides and therefore, we designed and customized generic interference oligonucleotides targeting both *AjCTP1* and *AjCTP2* (si*AjCTP1/2*-1 and si*AjCTP1/2*-2, containing the extra 2' ome

modification) (Beijing Tsingke Biotech Co., Ltd. China). Meanwhile, a negative control siRNA (siNC) was also customized for the experiment (Key resources table), which does not target genes in *A. japonicus*. The siAjCTP1/2-1, siAjCTP1/2-2 or siNC dry powder was diluted to 20 μ M with DEPC water following the manufacturer's instruction, then 60 μ l sterilized seawater was mixed with 20 μ l siAjCTP1/2-1, siAjCTP1/2-2 or siNC, and 20 μ l Lipo6000 transfection reagent (Beyotime, Shanghai, China) as the transfection solution. Eighteen sea cucumbers were randomly divided into three groups and injected with three different transfection solutions respectively. The mixed solutions were injected through the peristome once a day (1 μ l/g wet weight) and intestines were dissected and frozen in liquid nitrogen 3 days after injection. The expression level of AjCTP1/2 was investigated by qRT-PCR and the interference rate was calculated using $(1 - 2^{-\Delta\Delta CT})$ to determine the inhibition efficiency of siRNAs. Finally, the relative expression levels of AjCTR, AjPDRF1 and AjPDRF2 were investigated using the selected *A. japonicus* intestines.

To assess the effects of CT-type signaling inhibition on feeding and growth in *A. japonicus*, a 24-day RNAi experiment was conducted following the methodology outlined in [Section 4.8](#). Sea cucumbers were randomly allocated into two groups: the negative control group received siNC injections, while the experimental group was injected with siAjCTP1/2-1 (effectively targeting AjCTP1/2). Throughout the experiment, the dry weight of remaining bait and excrement, along with the wet weight of individual sea cucumbers, were recorded. Subsequently, WGR and SGR, as indices of feeding efficiency and growth performance, were calculated. In addition, the expression levels of the three growth factors (AjMegf6, AjIgf and AjGDF-8) in the intestinal tissue were tested by qPCR.

To investigate the potential mediation of CT-type signaling effects on feeding and growth by AjPDRF2 in *A. japonicus*, an RNAi experiment targeting AjPDRF2 was performed following the same knockdown protocol established for AjCTP. Briefly, the effective siRNA targeting AjPDRF2 (siAjPDRF2-1 or siAjPDRF2-2) was selected by testing AjPDRF2 expression in the intestines of injected sea cucumbers using qRT-PCR and calculating the interference rate efficiency. Subsequently, the validated siRNA was then employed in a 24-day RNAi experiment. The dry weight of remaining bait and excrement, as well as the wet weight of the sea cucumbers were measured, enabling calculation of WGR and SGR. Furthermore, the expression levels of three key growth factors (AjMegf6, AjIgf and AjGDF-8) in the intestinal tissue were analyzed via qPCR.

4.10 Statistical analysis

Statistical analyses were carried out using GraphPad Prism 8.0. Significant differences between groups were determined by one-way analysis of variance (ANOVA) followed by a Tukey's multiple comparisons test. Statistical significance thresholds were defined as follows: $p < 0.05$ (*) for significant differences and $p < 0.01$ (**) for highly significant differences, with annotations directly marked on the figures. All data were presented as mean $\pm$ SEM from at least three independent biological replicates. Non-linear regression analysis was performed using a four-parameter logistic equation with automatic outlier exclusion to fit the dose-response curve. The half maximal effective concentration (EC₅₀) was derived from the fitted curves through built-in algorithms in GraphPad Prism 8.0.

Data availability

All data generated or analyzed during this study are included in the manuscript and supporting files; source data files have been provided for all figures.

Additional information

Funding

This research was supported by National Natural Science Foundation of China [grant number 42276103] and Biotechnology and Biological Sciences Research Council (UK; grant numbers BB/M001644/1 and BB/X001024/1).

Author Contributions

Xiao Cong, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing—original draft preparation; Huachen Liu, Investigation, Methodology, Validation; Lihua Liu, Methodology, Validation; Nayeli Escudero-Castelán, Investigation, Methodology, Visualization, Writing—review & editing; Kite Jones, Investigation, Methodology, Visualization, Writing—review & editing; Michaela Egertová, Investigation, Methodology, Visualization; Maurice R. Elphick, Formal analysis, Funding acquisition, Resources, Supervision, Validation, Visualization, Writing—review & editing; Muyan Chen, Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing—review & editing

Funding

National Natural Science Foundation of China (42276103)

Biotechnology and Biological Sciences Research Council (BB/M001644/1)

Biotechnology and Biological Sciences Research Council (BB/X001024/1)

Supplementary figures



Figure 1-figure supplement 1.

Calcitonin-type (CT-type) neuropeptide precursors in *A. japonicus*.

(A) *AjCTP1* comprises two putative CT-type neuropeptides, AjCT1 and AjCT2, whereas *AjCTP1* generated by alternative splicing comprises only AjCT2. The predicted signal peptide is shown in blue, the predicted cleavage sites (KR or KK) are shown in green, CT-type neuropeptides are shown in red and C-terminal glycines that are potential substrates for amidation are shown in orange. Predicted disulfide bridges in AjCT1 and AjCT2 are formed between the two underlined cysteines. (B) Nucleotide sequences encoding CT-type neuropeptide precursors and the deduced amino acid sequences in *A. japonicus*. 3'UTR and 5'UTR are shown in lowercase and the ORF is shown in uppercase. Blue uppercase labels the predicted signal peptide, the green uppercase shows the predicted cleavage site, AjCT1 and AjCT2 are shown in red and the putative substrate for amidation is shown in orange.

A_japCT1 ---S- -S- -SNKFA- -G- -CAHMKV- -NAV L K Q N S R G Q Q Q F K F - -GSAG- -Pa
A_japCT2 ---RVGG- -GD- -FSG- -CAS L K A- -RDL V R A M L R - -PSKF- -GSGG- -Pa
H_scaCT1 ---S- -CSDRF- -S- -G- -CAHLKV- -AKAL L D Q - -ARREENSRF- -GISG- -Pa
H_scaCT2 ---MGG- -CGD- -FSG- -CAS L K A- -RDL V R A M L R Q - -PSKF- -GSGG- -Pa
H_glaCT1 ---S- -CSDRF- -S- -G- -CTHLKL- -AKVL I D Q - -SRRDGNIRF- -GSSG- -Pa
H_glaCT2 ---MGG- -CGDKF- -S- -G- -CAS L K A- -RDL V R A M L R Q - -PSKF- -GSGG- -Pa
S_purCT ---S- -KGG- -CGS- -FSG- -CMQMEV- -AKNRV A A L L R N S N A H L F - -CLNG- -Pa
A_rubCT ---NGESR- -G- -CGS- -F- -G- -CGVLT- -I- -CHNAA M R M L A E S - -NSPF- -GASG- -Pa
A_plaCT ---SSSG- -CAEYF- -G- -G- -CAQLKL- -GQDAL S R M L A D S - -NSRF- -GSGG- -Pa
P_pecCT ---S- -GTG- -CTQ- -FSG- -CAQLKV- -GQDAL S R V L A D S - -NSRF- -GSGG- -Pa
O_vicCT1 ---SGNGG- -CAG- -FTG- -CAQLAA- -GQNAL R N F M H S N R A S L F T - -GASG- -Pa
O_vicCT2 ---NGNGG- -CAG- -FTG- -CAQLAA- -GQSA L Q A M I H S G R A S L F - -GSGG- -Pa
O_araCT1 ---GTEKG- -CGS- -FSG- -CAQLAA- -GQSA L Q A M I H G N R A S L F - -GSGG- -Pa
O_araCT2 ---GGGG- -CKS- -FSG- -CAQLV- -GQNA V R N M M H S N R A S I F S - -GPRG- -Pa
A_filCT1 ---GTEKG- -CAG- -FSG- -CAQLAA- -GQSA L Q A M I H G N R A S L F - -GSGG- -Pa
A_filCT2 ---GGG- -CRG- -FSA- -CAQLA- -IGQDA F R N M M H N N R A S I F T - -SARG- -Pa
A_medCT ---G- -C- -DVF- -GG- -CAQLKV- -GRELA I D T L Q K G S G G M F - -GSSG- -Pa
An_japCT ---GDC- -DSF- -GG- -CAQLTA- -SQQLA Y R T R H R D L S N L F - -GTGG- -Pa
S_kowCT1 ---GEG- -CKG- -FAA- -CGNLD- -AGRKHWK A N M G G G R T A M E S S A S G T - -Pa
S_kowCT2 ---GRGTS- -AGG- -FAT- -CKQLEY- -GRKYA T S - -KADLSHF- -GATS- -Pa
B_floCT1 ---GK I A C K T - -AW- -CMNNRL- -SHNLS S L D N P T D T G V G - -A- -Pa
B_floCT2 ---DCST- -LT- -CFNQKL- -AHELA M D N Q R T D T A N P Y - -S- -Pa
B_floCT3 ---KCES- -GT- -CVQMHL- -ADRLR L G L G H N M F T N T - -GPES- -Pa
C_intCT ---CDG- -VST- -CWLHEL- -GNSVH A T A G G K Q - -NVGF- -G- -Pa
H_sapCGRP ---ACDT- -AT- -CVTHRL- -AGLLS R S G G V V K N N F V P T - -NVGSK A Fa
H_sapCT ---CGN- -LST- -CMLGTYTQDFNK F H T F P - -QTA I - -GVGA- -Pa
L_gigCT ---S- -CSLRL- -GGM- -CLTENL N A A A N Q Y - -EYLS- -SGLS- -Pa
D_melDH31 ---TVDFGL- -ARGYS- -G- -TQEAKHRMGLA- -AANF- -AGG- -Pa
C_telDH31 ---RFDAGY- -GS- -RYGV- -AQSVGSK L M A L K Q A A D W - -NG- -Pa
C_telCT ---T- -QFNLG- -GHC- -ATESA- -ASVADHW- -HYLN- -SPLS- -Pa

Figure 1-figure supplement 2.

Comparison of the sequences CT-type neuropeptides from Bilateria: deuterostomes-ambulacraria (green), deuterostomes-chordates (purple) and protostomes (pink).

The conserved residues are shown with white lettering highlighted in gray and black. The accession numbers or citations are shown in the [Supplementary table 1](#).

[illegible]

Figure 2-figure supplement 1.

Amino acid sequences and phosphorylation sites of AjCTR, AjPDFR1 and AjPDFR2 in *A. japonicus*.

(A), (B) and (C) show the cDNA sequences encoding the receptors and their deduced amino acid sequences. ORFs are shown in uppercase; 3'UTR and 5'UTR are shown in lowercase. The stop codon is shown by the asterisk (*). Predicted potential phosphorylation sites are shown with squares.

C

AjpPDFR2

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1  cgaggagctctctcttttgaagggtttttcaaaacggattgtgcttgtgtgtgacttc
61  aaagtccttgtagcgccaacagtttctagctgtcaacaatccttataggtaacgggtcacc
121  ttttgatgtgtctctcaacatgtgtgtctatttggctgtttacctaagccttttagag
181  caattcgagatctcttcaaggggggaaagttaaacggaggttcagaggagcctatactag
241  gggaaagatatacttaataatcagtgctagatatacaacacagttgtgtatggattcat
301  tagcgaagtgcctttttttgtcaatagaggaccgatcgtaaaagtatagaagaccaacgta
361  tatagtattaccacggttgagtgcatgtcggaagtgtcctatgtgacttaagctagtcgaag
421  acgtataccacggttatcatctgggatagtttggctcttcttctgtatcacgtatacat
481  ggaagaagaagcgttacttttagctgagcctgggtgattgtataaggagtgactgactct
541  tccaacaagtgaacatttagatttaacatttgaagacgagtggttataagaagcta
601  ggaagatcttgtgaagctogattgactacactgctgttttcaagagagttttacgcgta
1      M A N
661  gagttttgagtaacgtcggttcatactaaagttgatcttcagatcgtaagATGGCGAAC

4 N E T S V L C L E I C K D 5 D Y D V V C
721  AATGAAACAGGTGACTGTGTCTGGAATATGTAAGATTATGATTATGATGTAGTCTGT

24 W P E T A N G T I S N Q P C P D 5 P F T
781  TGGCCAGAAACAGCAAAACGGAACCATATCGAATCAGCCTTGTCTGACACGCTTTCACA

44 D P 5 Q L F E R L C Q P D G Q W E D V F
841  GACCCAAACAGCTCTTTGAGAGGTTGTGTAGCCTGATGGAATGGAAGACGTCTTC

64 M H 5 C I K G E F 5 R V T I P P 5 5 T E
901  ATGCATAGCTGATTAAAGAGAGATTTCTCGTGAACGATACCCCATCAAGTACGGAA

84 Q H D F L L F I V A I G R I I R L 5 G 5
961  CAACACGACTTTTCTGTTTATTGTAGGAGTTGGTGAATAATCCGGTTAACCGGTTC

104 L I 5 M L 5 L L L A L V I F A 5 F R 5 L
1021  TTGATATCTCGCTGAGTTTACTTCTCGCCCTTGTATCTTTCGCTCTTCAGGTCTTTG

124 K C N R T K I H V H L F T A L L I R L 5
1081  AAGTGCAACCGGACGAAGATCCAGTGCAATTATTACAGCGTTGTTGATTCGGCTCAC

144 I D V V F D I N N I L K Y K A G N D F R
1141  ATAGATGTCTGTGTTGATATTAAACAACATCTCAAATATAAGCGGGAATGATTTTCGA

164 A A D N I L G L C E A L H I V S T Y S 5
1201  GCAGCTGACAATATTCTGGGTTTATGTGAAGCGTTGCACATTGTTAGTACTTACGTAGC

184 L C A F A W M F V E G L 5 L N T L I 5 5
1261  CTCTGTGCGTTCGATGGATGTTTGTGGAAGGCTTGTACTGAACACGCTCATATCATCA

204 A V F G R P N F I R Y Y I I A W V L P I
1321  GCAGTATTTGGAAGACCAAAATTTATCGGTACTACATTATGTCATGGGTTCTCCGATC

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224 P F V L A W S I A M F A T Y S R C W P L
1381  CCGTTTGTGTAGCTTGGTCGATTGCAATGTTGCAACATATTCAAGATGCTGGGCTCTT

244 H 5 D 5 P Y Y L A L I E I P R D I I F I
1441  CATACTGACTCGCCCTACTACTTAGCACTAATAGAAATTCCTAGAGATATCATATTATT

264 L N 5 V F L L H I I Y V L V 5 K L K N N
1501  CTAATAGCGTGTTCCTTCTTCACATCATTTATGCTTGGTGACCAAAATAAAAATAAC

284 N 5 G E T Q V I R K A V K A L F V L L P
1561  AATTCTGGTGAACACAAGTCATAAGGAAGCAGTGAAGGCGCTGTTTGTCTCTCGCT

304 L L G I 5 H L L W R L P H P Q A 5 5 P P
1621  CTTCTGGGCATCAGCCATCTCTTTGGCGTCTTCTCATCGCAAGCTACAGGCCACCA

324 5 V I V I Y H V G I 5 F F Y S F Q G F A
1681  ACAGTCATTGTTATCTACACGTCGGCATATCGTCTTCTATTCGTTCCAGGATTTGCT

344 V A L L F C F L N K E V R V L I K R K W
1741  GTCGCTCTCTCTCTCTGTTTCTTAACAAGGAGGTTGCGTATTAAATAGAGGAAGTGG

364 R I W V N F M D 5 P M R R 5 5 M I T 5 5
1801  AGAATCTGGGTGAATTTTCATGGACACCGATGAGAAGGACTTCAATGATCAGCTCGCA

384 5 D V R I N L V M M N G K F N E *
1861  TCAGATGAAGGATTAACCTCGTCATGATGAACGGCAAAATTAATGAATGA

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Figure 2-figure supplement 1. (continued)

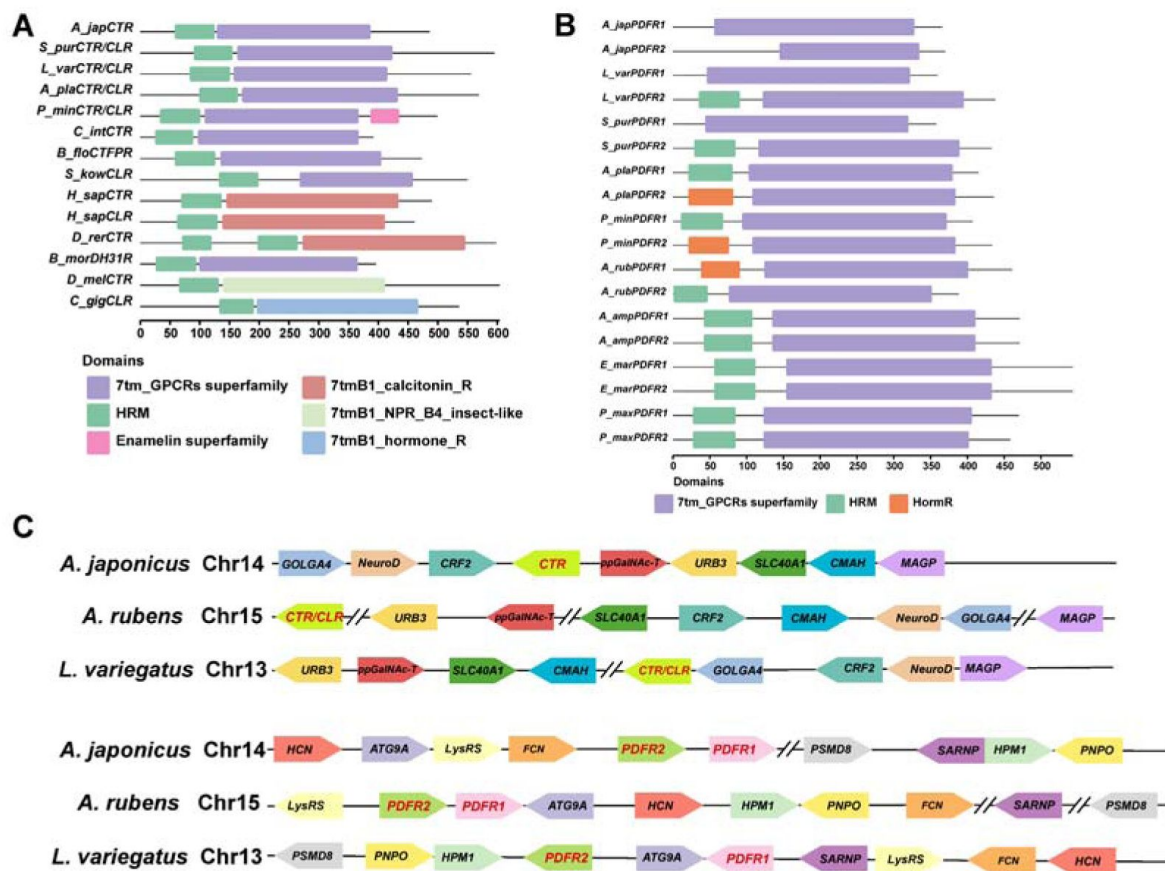


Figure 2-figure supplement 2.

Characterization of AjCT1 and AjCT2 receptors (AjCTR, AjPDR1 and AjPDR2).

(A) Conserved domains of CTR/CLR-type protein sequences. Black line represents the length of proteins and different domains are represented by rectangles. (B) Conserved domains of PDRs in Bilateria. Black line represents the length of proteins and different domains are represented by rectangles. (C) Synteny analysis of AjCT1 and AjCT2 receptors (AjCTR, AjPDR1 and AjPDR2). The chromosomal locations of AjCT1 and AjCT2 receptor genes from *A. rubens*, *L. variegatus* and *A. japonicus* are shown, with different colors used to mark neighboring genes and with their transcriptional direction shown by arrows. GOLGA4, golgin subfamily A member 4; NeuroD, neurogenic differentiation factor; CRF2, corticotropin-releasing factor receptor 2; CTR, calcitonin type receptor; CLR, calcitonin gene-related peptide type 1 receptor; ppGalNAc-T, polypeptide N-acetylgalactosaminyltransferase; URB3, E3 ubiquitin-protein ligase ubr3; SLC40A1, solute carrier family 40 member 1; CMAH, cytidine monophosphate-N-acetylneuraminic acid hydroxylase; MAGP, microfibril-associated glycoprotein; HCN, hyperpolarization-gated and cyclic nucleotide regulated channel; ATG9A, autophagy-related protein 9A-like; LysRS, lysine--tRNA ligase; FCN, ficolin; PDR1, pigment-dispersing factor receptor 1; PDR2, pigment-dispersing factor receptor 2; PSMD8, 26S proteasome regulatory subunit 8; SARNP, SAP domain-containing ribonucleoprotein; HPM1, histidine protein methyltransferase 1; PNPO, pyridoxine-5'-phosphate oxidase.

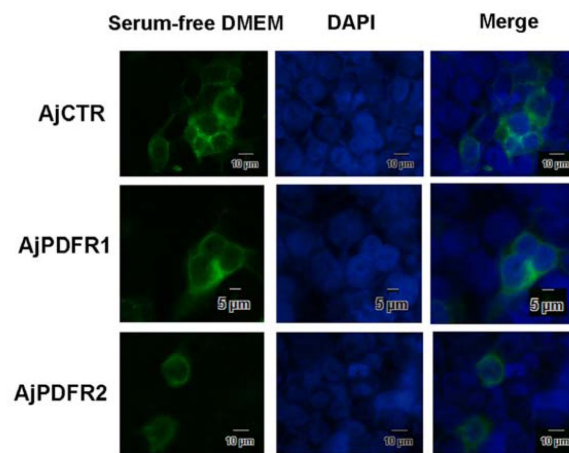
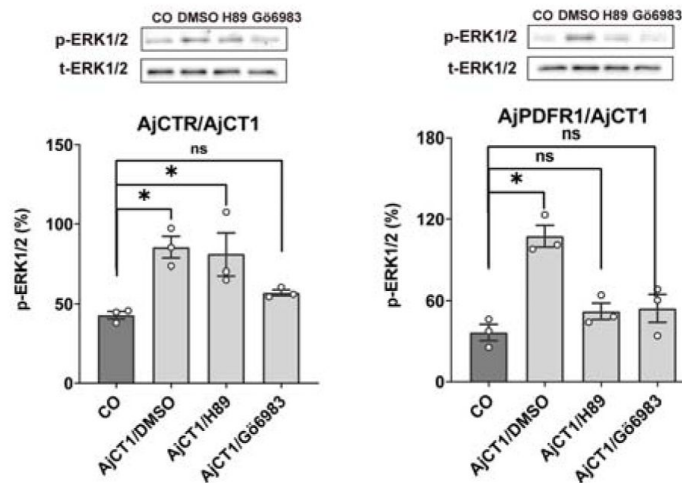


Figure 4-figure supplement 1.

Negative control for analysis of AjCTR, AjPDFR1 and AjPDFR2 internalization.

The images show the localization of AjCTR, AjPDFR1 and AjPDFR2 in HEK293T cells without AjCT1 or AjCT2 stimulation. Green fluorescence represents the localization of pEGFP-N1/receptors in cell membrane. Cell nucleus probe (DAPI) was applied for cell nucleus staining. All pictures are representative of three independent experiments.

A



B

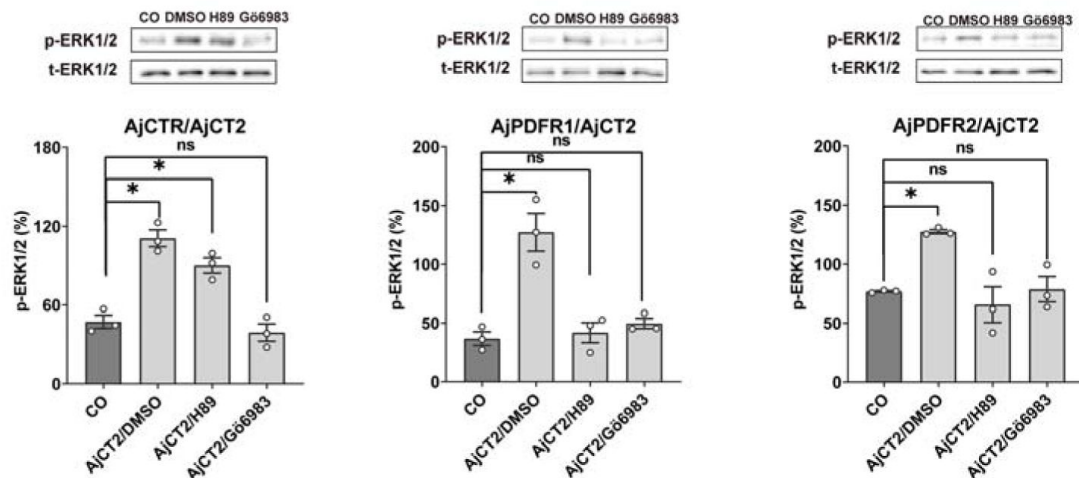


Figure 5-figure supplement 1.

Effects of PKA or PKC inhibitor on AjCT1 or AjCT2 stimulated ERK1/2 phosphorylation in AjCTR, AjPDFR1 or AjPDFR2 expressing HEK293T cells.

(A) ERK1/2 phosphorylation, activated by AjCT1, is blocked by PKC inhibitor Gö 6983 in AjCTR expressing HEK293T cells and is blocked by PKA inhibitor H89 and PKC inhibitor in AjPDFR1 expressing HEK293T cells. (B) ERK1/2 phosphorylation, activated by AjCT2, is blocked by PKC inhibitor in AjCTR expressing HEK293T cells and is blocked by PKA inhibitor and PKC inhibitor in AjPDFR1 and AjPDFR2 expressing HEK293T cells. Serum-starved HEK293 cells were pre-treated with DMSO, PKA inhibitor, or PKC inhibitor before AjCT1 or AjCT2 stimulation. All images are representative of three independent experiments. Error bars represent SEM for three independent experiments. Statistically significant differences are indicated as follows: *: $p < 0.05$, ns: no significant.

Figure 7-figure supplement 1

Immunostaining in the radial nerve cord of *A. japonicus* is abolished by pre-absorption of ArCT antiserum with the ArCT antigen peptide.

(A) Transverse section of a juvenile specimen of *A. japonicus* showing immunostaining in the radial nerve cord revealed by an antiserum to the starfish calcitonin-type neuropeptide ArCT, as also shown in Figure 7A. (B) A transverse section neighbouring the section shown in A was incubated with ArCT antiserum that had been pre-absorbed with the ArCT antigen peptide; note the absence of immunostaining in the radial nerve cord, demonstrating the specificity of the immunostaining in A. Abbreviations: LM, longitudinal muscle; RNC, radial nerve cord. Scale is 100 μm .

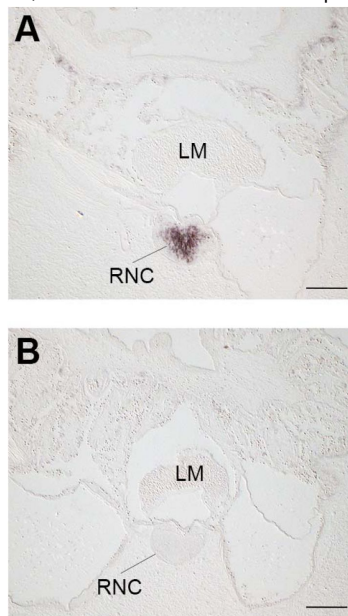


Figure 8-figure supplement 1.

Positive control on experimental tissues and pharmacological effects of 10^{-6} M AjCT2 on longitudinal muscle from *A. japonicus*.

(A) shows a recording of the relaxing effect of AjCT2 (10^{-6} M) on a longitudinal muscle preparation. (B) and (C) respectively are recordings of the contracting effects of ACh on longitudinal muscle and intestine preparations, showing that the effect of ACh persists for at least 15 minutes. All recordings shown are representative of three independent experiments.

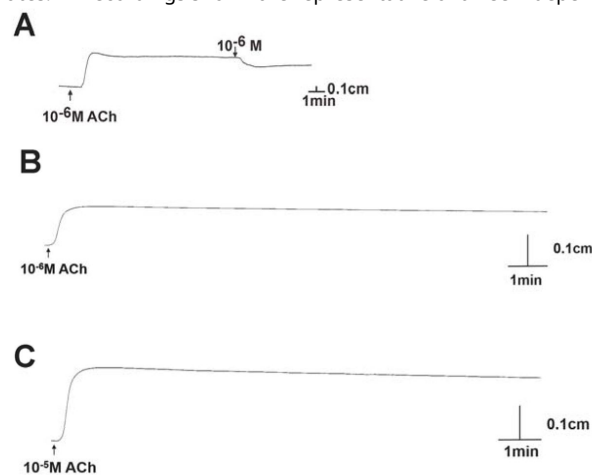


Figure 9A-figure supplement 1.

Measurement of the mass of remaining bait and excrement for AjCT2-treated and control animals.

(A) Remaining bait mass in different phases. (B) The mass of excrement in different phases. The 24-day experiment was divided into six phases (I-VI), with each phase comprising four days for statistical analysis. CO: control group (sterilized seawater); CT2L: AjCT2 low concentration group (5×10^{-3} mg/mL); CT2H: AjCT2 high concentration group (5×10^{-1} mg/mL). Values are means $\pm$ SEM (n = 3, representing three parallel experiments of each group). Asterisks represent the significant difference as follows: *: $p < 0.05$, **: $p < 0.01$.

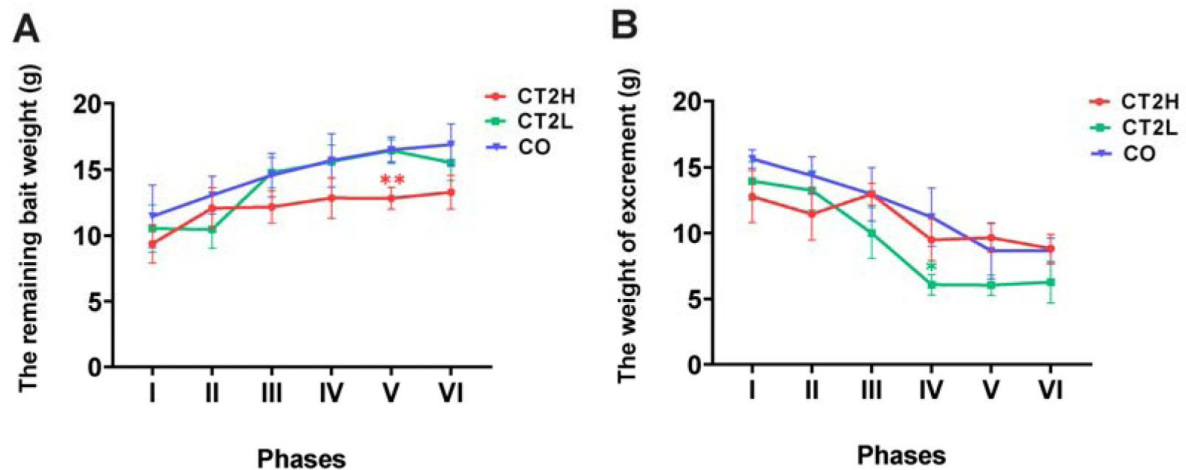
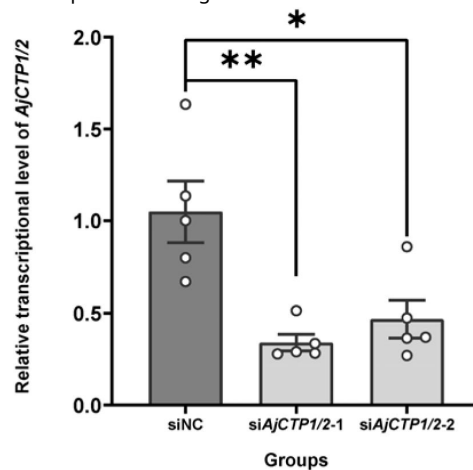


Figure 10-figure supplement 1.

The relative expression level of *AjCTP1/2* in siNC, si*AjCTP1/2-1* and si*AjCTP1/2-2*. siNC: negative control group; si*AjCTP1/2-1* and si*AjCTP1/2-2*: experimental groups targeting *AjCTP1/2*.

Values are means $\pm$ SEM (n = 5). Asterisks represent the significant differences as follows: *: $p < 0.05$, **: $p < 0.01$.



Supplementary tables

Abbreviation	Species Names	Accession Numbers And/or Citations
<i>H_sapCTP</i>	<i>Homo sapiens</i>	GI:179819
<i>H_sapCGRPP</i>	<i>Homo sapiens</i>	NP_001029125.1
<i>T_rubCTP</i>	<i>Takifugu rubripes</i>	GenBank: CAC81278.1
<i>T_rubCGRPP</i>	<i>Takifugu rubripes</i>	GenBank: CAC81277.1
<i>C_intCTP</i>	<i>Ciona intestinalis</i>	GenBank: BAI63095.1
<i>B_floCTP1</i>	<i>Branchiostome floridae</i>	GI:260826569
<i>B_floCTP2</i>	<i>Branchiostome floridae</i>	GI:260826567
<i>B_floCTP3</i>	<i>Branchiostome floridae</i>	GI:260826573
<i>S_kowCTP</i>	<i>Saccoglossus kowalevskii</i>	GI:187217193
<i>A_japCTP1</i>	<i>Apostichopus japonicus</i>	GenBank: AWU78769.1
<i>A_japCTP2</i>	<i>Apostichopus japonicus</i>	GenBank: AWU78770.1
<i>A_rubCTP</i>	<i>Asterias rubens</i>	GenBank: KT601715.1
<i>P_pecCTP</i>	<i>Patiria pectinifera</i>	GenBank: AWW63418.1
<i>A_plaCTP</i>	<i>Acanthaster planci</i>	Smith et al. (2017)
<i>S_purCTP</i>	<i>Strongylocentrotus purpuratus</i>	GI:115767208
<i>O_vicCTP1</i>	<i>Ophionotus. victoriae</i>	GenBank: ASK86238.1
<i>O_vicCTP2</i>	<i>Ophionotus. victoriae</i>	GenBank: ASK86238.1
<i>O_araCTP1</i>	<i>Ophiopsila aranea</i>	Zandawala et al. (2017)
<i>O_araCTP2</i>	<i>Ophiopsila aranea</i>	
<i>A_filCTP1</i>	<i>Amphiura filiformis</i>	
<i>A_filCTP2</i>	<i>Amphiura filiformis</i>	
<i>H_scaCTP1</i>	<i>Holothuria scabra</i>	Suwansa-Ard et al. (2018)
<i>H_scaCTP2</i>	<i>Holothuria scabra</i>	
<i>H_glaCTP1</i>	<i>Holothuria glaberrima</i>	
<i>H_glaCTP2</i>	<i>Holothuria glaberrima</i>	
<i>A_medCTP</i>	<i>Antedon mediterranea</i>	Aleotti et al., (2022)
<i>An_japCTP</i>	<i>Anneissia japonica</i>	
<i>L_gigCTP</i>	<i>Lottia gigantea</i>	GI:163526287
<i>D_melDH31P</i>	<i>Drosophila melanogaster</i>	NP_523514.1
<i>L_migDH31P</i>	<i>Locusta migratoria</i>	GenBank: AKN21237.1
<i>C_telCTP</i>	<i>Capitella teleta</i>	GI: 161220966
<i>C.telDH31P</i>	<i>Capitella teleta</i>	Rowe et al. (2014)

Supplementary table 1.

Accession numbers or citations of calcitonin-type family peptides or precursors in Bilateria.

Species	Protein ID	Gene code	Species code
<i>Amphiura filiformis</i>	XP_072029112.1	CRHR	Afil
<i>Amphiura filiformis</i>	XP_072021544.1	CTR	Afil
<i>Amphiura filiformis</i>	XP_072029557.1	PDFR1	Afil
<i>Amphiura filiformis</i>	XP_072028436.1	PDFR2	Afil
<i>Anneissia japonica</i>	XP_033108182.1	CRHR	Anja
<i>Anneissia japonica</i>	XP_033104142.1	CTR	Anja
<i>Anneissia japonica</i>	XP_033104113.1	PDFR	Anja
<i>Apis mellifera</i>	XP_006564466.1	DH31R	Amel
<i>Apis mellifera</i>	XP_016768833.2	PDFR	Amel
<i>Aplysia californica</i>	XP_012938918.2	CRHR	Acal
<i>Aplysia californica</i>	XP_005089839.2	CTR	Acal
<i>Aplysia californica</i>	XP_005093780.3	DH31R	Acal
<i>Apostichopus japonicus</i>	XP_071811976.1	CRHR	Ajap
<i>Apostichopus japonicus</i>	PIK36177.1	CRHR	Ajap
<i>Apostichopus japonicus</i>	XP_071811994.1	CRHR	Ajap
<i>Apostichopus japonicus</i>	PIK41405.1	CRHR	Ajap
<i>Apostichopus japonicus</i>	XP_071812618.1	CTR	Ajap
<i>Apostichopus japonicus</i>	XP_071811731.1	PDFR1	Ajap
<i>Apostichopus japonicus</i>	XP_071812243.1	PDFR2	Ajap
<i>Asterias rubens</i>	ENSASRP00000008640.1	CRHR	Arub
<i>Asterias rubens</i>	ENSASRP00000013963.1	CRHR	Arub
<i>Asterias rubens</i>	ENSASRP00000014061.1	CRHR	Arub
<i>Asterias rubens</i>	ENSASRP00000011186.1	CTR	Arub
<i>Asterias rubens</i>	ENSASRP00000010309.1	PDFR1	Arub
<i>Asterias rubens</i>	ENSASRP00000010166.1	PDFR2	Arub
<i>Bombus impatiens</i>	ENSBIMP00000009704.1	CRHR	Bimp
<i>Bombus impatiens</i>	ENSBIMP00000023653.1	DH31R	Bimp
<i>Bombus impatiens</i>	ENSBIMP00000026387.1	PDFR	Bimp
<i>Branchiostoma floridae</i>	XP_035687893.1	CRHR	Bflo
<i>Branchiostoma floridae</i>	XP_035687894.1	CRHR	Bflo
<i>Branchiostoma floridae</i>	XP_035689670.1	CRHR	Bflo
<i>Branchiostoma floridae</i>	XP_035689671.1	CRHR	Bflo
<i>Branchiostoma floridae</i>	XP_035687895.1	CTR*	Bflo
<i>Caenorhabditis elegans</i>	NP_001122749.1	SEB2*	Cele
<i>Ciona intestinalis</i>	XP_002122381.1	CRHR	Cint
<i>Ciona intestinalis</i>	XP_009858716.1	CRHR	Cint
<i>Drosophila melanogaster</i>	NP_725278.1	DH31R*	Dmel

<i>Drosophila melanogaster</i>	NP_001285174.1	HECTOR	Dmel
<i>Drosophila melanogaster</i>	CAB72288.1	PDFR	Dmel
<i>Gallus gallus</i>	ENSGALP00010018855.1	CLR*	Ggal
<i>Gallus gallus</i>	ENSGALP00010039362.1	CRHR	Ggal
<i>Gallus gallus</i>	ENSGALP00010039367.1	CRHR	Ggal
<i>Gallus gallus</i>	ENSGALP00010033414.1	CRHR	Ggal
<i>Gallus gallus</i>	ENSGALP00010004560.1	CTR*	Ggal
<i>Gekko gecko</i>	ENSEGEP00000030211.1	CLR	Ggec
<i>Gekko gecko</i>	ENSEGEP00000005175.1	CRHR	Ggec
<i>Gekko gecko</i>	ENSEGEP00000009950.1	CRHR	Ggec
<i>Gekko gecko</i>	ENSEGEP00000001898.1	CTR	Ggec
<i>Homo sapiens</i>	ENST04850112052.1	CLR*	Hsap
<i>Homo sapiens</i>	ENST04850047995.1	CRHR	Hsap
<i>Homo sapiens</i>	ENST04850093643.1	CRHR	Hsap
<i>Homo sapiens</i>	ENST04850217190.1	CTR*	Hsap
<i>Hypsibius exemplaris</i>	OQV19025.1	CRHR	Hexe
<i>Hypsibius exemplaris</i>	OQV18662.1	CRHR	Hexe
<i>Hypsibius exemplaris</i>	OWA52582.1	CTR	Hexe
<i>Hypsibius exemplaris</i>	OWA52579.1	DH31Ra	Hexe
<i>Hypsibius exemplaris</i>	OQV24416.1	DH31Rb	Hexe
<i>Hypsibius exemplaris</i>	OQV13187.1	PDFR	Hexe
<i>Lepisosteus oculatus</i>	XP_015199664.1	CLRa	Locu
<i>Lepisosteus oculatus</i>	XP_015214520.1	CLRb	Locu
<i>Lepisosteus oculatus</i>	XP_015209811.1	CRHR	Locu
<i>Lepisosteus oculatus</i>	XP_015217722.1	CRHR	Locu
<i>Lepisosteus oculatus</i>	XP_006636078.2	CTR	Locu
<i>Limulus polyphemus</i>	XP_013771881.1	CRHR	Lpol
<i>Limulus polyphemus</i>	XP_022240360.1	CRHR	Lpol
<i>Limulus polyphemus</i>	XP_013775447.1	CRHR	Lpol
<i>Limulus polyphemus</i>	XP_013791145.1	CRHR	Lpol
<i>Limulus polyphemus</i>	XP_013791446.1	CRHR	Lpol
<i>Limulus polyphemus</i>	XP_022240480.1	CTRa	Lpol
<i>Limulus polyphemus</i>	XP_022240470.1	CTRb	Lpol
<i>Limulus polyphemus</i>	XP_022254242.1	CTRc	Lpol
<i>Limulus polyphemus</i>	XP_022237602.1	CTRd	Lpol
<i>Limulus polyphemus</i>	XP_022246293.1	DH31Ra	Lpol
<i>Limulus polyphemus</i>	XP_022235154.1	DH31Rb	Lpol
<i>Limulus polyphemus</i>	XP_022235431.1	DH31Rc	Lpol
<i>Limulus polyphemus</i>	XP_022239928.1	DH31Rd	Lpol
<i>Limulus polyphemus</i>	XP_022248891.1	DH31Re	Lpol
<i>Limulus polyphemus</i>	XP_022255301.1	PDFR	Lpol
<i>Lineus longissimus</i>	ENSLNLP00015009405.1	CRHR	Llon
<i>Lineus longissimus</i>	ENSLNLP00015009422.1	CRHR	Llon
<i>Lineus longissimus</i>	ENSLNLP00015009434.1	CRHR	Llon
<i>Lineus longissimus</i>	ENSLNLP00015009638.1	CTRa	Llon
<i>Lineus longissimus</i>	ENSLNLP00015008640.1	CTRb	Llon
<i>Lineus longissimus</i>	ENSLNLP00015012550.1	CTRc	Llon
<i>Lineus longissimus</i>	ENSLNLP00015013093.1	DH31R	Llon
<i>Lineus longissimus</i>	ENSLNLP00015009635.1	PDFR	Llon
<i>Lumbricus terrestris</i>	ENSVPPP00000022107.1	CTRa	Lter

<i>Lumbricus terrestris</i>	ENSVPPP00000011858.1	CTRb	Lter
<i>Lumbricus terrestris</i>	ENSVPPP00000018690.1	CTRc	Lter
<i>Lumbricus terrestris</i>	ENSVPPP00000033159.1	CTRd	Lter
<i>Lumbricus terrestris</i>	ENSVPPP00000030225.1	DH31Ra	Lter
<i>Lumbricus terrestris</i>	ENSVPPP00000022149.1	DH31Rb	Lter
<i>Lumbricus terrestris</i>	ENSVPPP00000003533.1	PDFR	Lter
<i>Magallana gigas</i>	NP_001292279.1	CTR1*	Mgig
<i>Magallana gigas</i>	XP_034310675.2	CTR2*	Mgig
<i>Magallana gigas</i>	XP_011442634.2	CTR3	Mgig
<i>Magallana gigas</i>	XP_011442636.3	CTR4	Mgig
<i>Magallana gigas</i>	XP_065929766.1	CTR5	Mgig
<i>Magallana gigas</i>	XP_065923133.1	CTR6	Mgig
<i>Magallana gigas</i>	XP_052684037.1	CTR7	Mgig
<i>Magallana gigas</i>	XP_034317823.1	PDFR	Mgig
<i>Magallana gigas</i>	XP_034310400.1		Mgig
<i>Owenia fusiformis</i>	CAC9666549.1	CRHR	Ofus
<i>Owenia fusiformis</i>	CAC9666335.1	CTRa	Ofus
<i>Owenia fusiformis</i>	CAC9673671.1	CTRb	Ofus
<i>Owenia fusiformis</i>	CAC9666125.1	DH31R	Ofus
<i>Owenia fusiformis</i>	CAC9578850.1	PDFR	Ofus
<i>Paramacrobrotus metropolita</i>	XP_055352663.1	CRHR	Pmet
<i>Paramacrobrotus metropolita</i>	XP_055352664.1	CRHR	Pmet
<i>Paramacrobrotus metropolita</i>	XP_055329231.1	CTR	Pmet
<i>Paramacrobrotus metropolita</i>	XP_055329233.1	DH31R	Pmet
<i>Paramacrobrotus metropolita</i>	XP_055331530.1	PDFR	Pmet
<i>Pecten maximus</i>	XP_033762144.1	CRHR	Pmax
<i>Pecten maximus</i>	XP_033762006.1	CTRa	Pmax
<i>Pecten maximus</i>	XP_033762916.1	CTRb	Pmax
<i>Pecten maximus</i>	XP_033762874.1	CTRc	Pmax
<i>Pecten maximus</i>	XP_033762263.1	CTRd	Pmax
<i>Pecten maximus</i>	XP_033762155.1	DH31Ra	Pmax
<i>Pecten maximus</i>	XP_033762946.1	DH31Rb	Pmax
<i>Pecten maximus</i>	XP_033729115.1	DH31Rc	Pmax
<i>Pecten maximus</i>	XP_033762294.1	PDFR	Pmax
<i>Platynereis dumerilii</i>	AKQ63006.1	DH31R-1*	Pdum
<i>Platynereis dumerilii</i>	WOE99345.1	PDFR	Pdum
<i>Priapulid caudatus</i>	XP_014664111.1	DH31Ra	Pcau
<i>Priapulid caudatus</i>	XP_014661438.1	DH31Rb	Pcau
<i>Rotaria socialis</i>	CAF3356201.1	PDFRa	Rsoc
<i>Rotaria socialis</i>	CAF3546535.1	PDFRb	Rsoc
<i>Saccoglossus kowalevskii</i>	NP_001161520.1	CRHR	Skow
<i>Saccoglossus kowalevskii</i>	XP_006818813.1	CTR	Skow
<i>Saccoglossus kowalevskii</i>	XP_006825104.1	PDFRa	Skow
<i>Saccoglossus kowalevskii</i>	XP_006825103.1	PDFRb	Skow
<i>Schmidtea mediterranea</i>	SMEST051050001	CRHR	Smed
<i>Schmidtea mediterranea</i>	SMEST064388001	CRHR	Smed
<i>Schmidtea mediterranea</i>	SMEST049048003	CRHR	Smed
<i>Schmidtea mediterranea</i>	SMEST080399003	CTRa	Smed
<i>Schmidtea mediterranea</i>	SMEST042095002	CTRb	Smed
<i>Schmidtea mediterranea</i>	SMEST006966001	DH31Ra	Smed

<i>Schmidtea mediterranea</i>	SMEST027375002	DH31Rb	Smed
<i>Schmidtea mediterranea</i>	SMEST023013001	DH31Rc	Smed
<i>Schmidtea mediterranea</i>	SMEST035956002	PDFR	Smed
<i>Sepioteuthis lessoniana</i>	ENSUKOP00000023564.1	CRHR	Sles
<i>Sepioteuthis lessoniana</i>	ENSUKOP00000009609.1	CTRa	Sles
<i>Sepioteuthis lessoniana</i>	ENSUKOP00000023956.1	CTRc	Sles
<i>Sepioteuthis lessoniana</i>	ENSUKOP00000023961.1	CTRd	Sles
<i>Sepioteuthis lessoniana</i>	ENSUKOP00000005365.1	DH31Ra	Sles
<i>Sepioteuthis lessoniana</i>	ENSUKOP00000007960.1	DH31Rb	Sles
<i>Sepioteuthis lessoniana</i>	ENSUKOP00000030254.1	PDFR	Sles
<i>Strongylocentrotus purpuratus</i>	XP_003725174.2	CRHR	Spur
<i>Strongylocentrotus purpuratus</i>	XP_030845739.1	CRHR	Spur
<i>Strongylocentrotus purpuratus</i>	XP_030846612.1	CRHR	Spur
<i>Strongylocentrotus purpuratus</i>	XP_030846621.1	CTR	Spur
<i>Strongylocentrotus purpuratus</i>	XP_030845952.1	PDFR1	Spur
<i>Strongylocentrotus purpuratus</i>	XP_030845953.1	PDFR2	Spur
<i>Styela clava</i>	XP_039255657.1	CRHR	Scla
<i>Tribolium castaneum</i>	XP_008198347.1	CRHR	Tcas
<i>Tribolium castaneum</i>	XP_008192711.1	CRHR	Tcas
<i>Tribolium castaneum</i>	XP_064212169.1	CTR	Tcas
<i>Tribolium castaneum</i>	XP_008193920.1	DH31Ra	Tcas
<i>Tribolium castaneum</i>	XP_064210831.1	DH31Rb	Tcas
<i>Tribolium castaneum</i>	XP_064212793.1	PDFR	Tcas
<i>Xenopus tropicalis</i>	ENSXETP00000114756.1	CLR	Xtro
<i>Xenopus tropicalis</i>	ENSXETP00000109962.1	CRHR	Xtro
<i>Xenopus tropicalis</i>	ENSXETP00000101014.1	CRHR	Xtro
<i>Xenopus tropicalis</i>	ENSXETP00000060302.3	CTR	Xtro

Supplementary table 2.

Accession numbers of CT-type, PDF-type and CRH-type receptor families in Bilateria.

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Reviewer #1 (Public review):

Summary:

The manuscript characterizes a functional peptidergic system in the echinoderm *Apostichopus japonicus* that is related to the widely conserved family of calcitonin/diuretic hormone 31 (CT/DH31) peptides in bilaterian animals. In vitro analysis of receptor-ligand interactions, using multiple receptor activation assays, identifies three cognate receptors for two CT-like peptides in the sea cucumber, which stimulate cAMP, calcium, and ERK signaling. Only one of these receptors clusters within the family of calcitonin and calcitonin-like receptors (CTR/CLR) in bilaterian animals, whereas two other receptors cluster with invertebrate pigment dispersing factor receptors (PDFRs). In addition, this study sheds light on the expression and in vivo functions of CT-like peptides in *A. japonicus*, by quantitative real-time PCR, immunohistochemistry, pharmacological experiments on body wall muscle and intestine preparations, and peptide injection and RNAi knockdown experiments. This reveals a conserved function of CT-like peptides as muscle relaxants and growth regulators in *A. japonicus*.

Strengths:

This work combines both in vitro and in vivo functional assays to identify a CT-like peptidergic system in an economically relevant echinoderm species, the sea cucumber *A. japonicus*. A major strength of the study is that it identifies three G protein-coupled receptors for AjCT-like peptides, one related to the CTR/CLR family and two related to the PDFR family. A similar finding was previously reported for the CT-related peptide DH31 in *Drosophila melanogaster* that activates both CT-type and PDF-type receptors. Here, the authors expand this observation to a deuterostomian animal, which suggests that receptor promiscuity is a more general feature of the CT/DH31 peptide family and that CT/DH31-like peptides may activate both CT-type and PDF-type receptors in other animals as well.

Besides the identification of receptor-ligand pairs, the downstream signaling pathways of AjCT receptors have been characterized, revealing broad and in some cases receptor-specific effects on cAMP, calcium, and ERK signaling.

Functional characterization of the CT-related peptide system in heterologous cells is complemented with ex vivo and in vivo experiments. First, peptide injection and RNAi knockdown experiments establish transcriptional regulation of all three identified receptors in response to changing AjCT peptide levels. Second, ex vivo experiments reveal a conserved role for the two CT-like peptides as muscle relaxants, which have differential effects on body wall muscle and intestine preparations. Finally, peptide injection and knockdown experiments uncover a growth-promoting role for one CT-like peptide (AjCT2). Injection of AjCT2 at high concentration, or long-term knockdown of the AjCT precursor, affects diverse growth-related parameters including weight gain rate, specific growth rate, and transcript levels of growth-regulating transcription factors. The authors also reveal a growth-promoting function for the PDFR-like receptor AjPDFR2, suggesting that this receptor mediates the effects of AjCT2 on growth.

Weaknesses:

Expression of CT-like peptides was investigated both at transcript and protein level, but insight into the expression of the three peptide receptors is limited. This makes it difficult to understand the mechanism underlying the (different) functions of the two CT-like peptides in vivo. The authors identify differences in signal transduction cascades activated by each peptide, which might underpin distinct functions, but these differences were established only in heterologous cells.

The authors show overlapping phenotypes for a long-term knockdown of the AjCT precursor and the AjPDFR2 receptor, suggesting that the growth-regulating functions of AjCT2 are mediated by this receptor pathway. However, it remains unclear whether this mechanism underpins the growth-regulating function of AjCT2, until further in vivo evidence for this ligand-receptor interaction is presented. For example, the authors could investigate whether knockdown of AjPDFR2 attenuates the effects of AjCT2 peptide injection. In addition, a functional PDF system in this species remains uncharacterized, and a potential role of PDF-like peptides in growth regulation has not yet been investigated in *A. japonicus*. Therefore, it also remains unclear whether the ability of CT-like peptides to activate PDFRs is an evolutionary ancient property of this peptide family or whether this is an example of convergent evolution in some protostomian (*Drosophila*) and deuterostomian (sea cucumber) species.

<https://doi.org/10.7554/eLife.101799.3.sa2>

Reviewer #2 (Public review):**Summary:**

The authors show that *A. japonicus* calcitonins (AjCT1 and AjCT2) activate not only the calcitonin/calcitonin-like receptor, but they also activate the two "PDF receptors", *ex vivo*. They also explore secondary messenger pathways that are recruited following receptor activation. They determine the source of CT1 and CT2 using qPCR and *in situ* hybridization and finally test the effects of these peptides on tissue contractions, feeding and growth. This study provides solid evidence that CT1 and CT2 act as ligands for calcitonin receptors; however, evidence supporting cross-talk between CT peptides and "PDF receptors" is weak.

Strengths:

This is the first study to report pharmacological characterization of CT receptors in an echinoderm. Multiple lines of evidence in cell culture (receptor internalization and secondary messenger pathways) support this conclusion.

<https://doi.org/10.7554/eLife.101799.3.sa1>

Author response:

The following is the authors' response to the previous reviews

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Weaknesses:

The authors present a more detailed phylogenetic analysis in the revised version, including a larger number of species. But some clusters in the analysis are not well supported because they have only low bootstrap values. This makes it difficult to interpret the clustering in some parts of the tree.

Thank you for the reviewer's comments. In response, we have produced a new phylogenetic analysis using the maximum likelihood method. This was done by Nayeli Escudero Castela and Kite Jones in the Elphick group at QMUL and therefore they have been added as co-authors of this paper. The new phylogenetic tree (Figure 2, line 206) includes broad taxonomic sampling of CT-type receptors and PDF-type receptors. CRH-type receptors, which are also members of the secretin-type GPCR sub-family, have been included as an outgroup to root the tree. In the previous version the much more distantly related vasopressin/oxytocin-type receptors, which are rhodopsin-type GPCRs, were included as an outgroup. Furthermore, VIP-type receptors were also included in the previous tree but these have been omitted from the new tree because VIP receptor orthologs only occur in vertebrates and therefore they are not representative of a bilaterian GPCR family. The new tree shows high bootstrap support for key clades, notably achieving a bootstrap value of 100 for a clade comprising both deuterostomian and protostomian PDF receptors. This provides important evidence that the *A. japonicus* PDF-type receptors characterised in this study (AjPDFR1, AjPDFR2) are co-orthologs of the PDF-type receptor that has been characterised previously in *Drosophila*. Similarly, there is strong bootstrap support (100) for a clade comprising CT/DH31-type receptors and, importantly, the CT-type receptor characterised in this study (AjCTR) is positioned in a branch of this clade that comprises deuterostomian CT-type receptors (with bootstrap support of 100). Details of methods employed to produce the new receptor tree are included in lines 727-739. The new phylogenetic tree is shown below and has been incorporated into the revised manuscript (Figure 2, line 206). The description of new phylogenetic tree has also been modified accordingly in the revised manuscript (line 169-183).

References:

- Bauknecht P, Jékely G. Large-Scale Combinatorial Deorphanization of Platynereis Neuropeptide GPCRs. *Cell reports*, 2015, 12(4), 684–693. doi: 10.1016/j.celrep.2015.06.052.
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Expression of CT-like peptides was investigated both at transcript and protein level, but insight into the expression of the three peptide receptors is limited. This makes it difficult to understand the mechanism underlying the (different) functions of the two CT-like peptides in vivo. The authors identify differences in signal transduction cascades activated by each peptide, which might underpin distinct functions, but these differences were established only in heterologous cells.

We appreciate the reviewer's insightful comments. Regarding expression of CT-like peptide receptors, we have quantitatively analyzed the mRNA expression levels of the three receptors in key tissues using qRT-PCR (Figure 6, line 319) and receptor expression exhibits significant tissue-specific differences. Combined with the heterologous expression assays and In vivo functional validation, we believe our findings have provided clear mechanistic insights into the functional divergence of the two CT-like peptides. Investigation of the expression of the three receptor proteins in *A. japonicus* would require generation of specific antibodies, which was beyond the scope of this study. Furthermore, immunohistochemical visualization of neuropeptide receptor expression in other invertebrates has not been reported widely, which likely reflects technical difficulties in generation of antibodies that can be used to specifically detect receptor proteins that are typically expressed a low level in comparison to the neuropeptides that act as their ligands.

We acknowledge that investigating signal transduction cascades in heterologous cells (rather than native *A. japonicus* cells) is a limitation. However, as a non-model organism, *A. japonicus* currently lacks established cell lines for such research. Therefore, using heterologous cells was the most feasible approach to examine the differential signaling cascades activated by the peptides through the three receptors. Importantly, our in vivo experiments demonstrated that long-term knockdown of either the AjCT precursor or AjPDFR2 resulted in similar and significant growth defects. The phenotypic consistency

strongly suggests that AjCT2 and AjPDFR2 function within the same signaling pathway, with AjPDFR2 serving as the key receptor functionally activated by AjCT2.

The authors show overlapping phenotypes for a long-term knockdown of the AjCT precursor and the AjPDFR2 receptor, suggesting that the growth-regulating functions of AjCT2 are mediated by this receptor pathway. However, it remains unclear whether this mechanism underpins the growth-regulating function of AjCT2, until further in vivo evidence for this ligand-receptor interaction is presented. For example, the authors could investigate whether knockdown of AjPDFR2 attenuates the effects of AjCT2 peptide injection. In addition, a functional PDF system in this species remains uncharacterized, and a potential role of PDF-like peptides in growth regulation has not yet been investigated in A. japonicus. Therefore, it also remains unclear whether the ability of CT-like peptides to activate PDFRs is an evolutionary ancient property of this peptide family or whether this is an example of convergent evolution in some protostomian (Drosophila) and deuterostomian (sea cucumber) species.

Thank you for the reviewer's insightful comments and constructive questions. We acknowledge the request for more direct evidence to demonstrate how AjCT2 functions in vivo through AjPDFR2. However, long-term knockdown of the AjCT precursor and AjPDFR2 both resulted in identical and significant growth defect phenotypes. The high phenotypic consistency, combined with the activation effect of AjCT2 on AjPDFR2 in heterologous cells, strongly suggests that they function within the same signaling pathway, with AjPDFR2 serving as the key receptor functionally activated by AjCT2. While exogenous peptide injection combined with receptor knockdown is a classic method for verifying receptor activation, phenotypic overlap itself is widely accepted in genetic research as robust evidence for pathway association (Shafer and Taghert, 2009; Van Sinay et al., 2017). A. japonicus is a non-model organism with a 3-month aestivation period in summer followed shortly by winter hibernation. During these periods, we are unable to conduct in vivo experiments. Any single experimental suggestion from reviewers could potentially require one more year of research and we have already conducted an additional year of research, in response to reviewer feedback, since submitting the original manuscript. We hope therefore that these challenges associated with working with aquatic invertebrate non-model organisms is recognized by the reviewers.

We fully agree that the functional PDF/PDFR system in A. japonicus and its potential role in growth regulation remain uncharacterized. Currently, the precursors of the PDF-type neuropeptide in echinoderms remain unidentified, which precludes clear pharmacological characterization of the two receptors. While further exploration of echinoderm PDF-type neuropeptides is still needed, our phylogenetic analysis-conducted using the maximum likelihood method with optimized parameters and rigorous sequence curation-demonstrates that the deuterostomian PDFRs (including AjPDFR1 and AjPDFR2) are positioned in a clade with the well-characterized protostomian PDFR clades with extremely high bootstrap support (value=100). Therefore, these two receptors in A. japonicus clearly belong to the PDF receptor family and our findings clearly indicate that the ability of CT-like peptides to activate PDFRs is either an evolutionarily ancient and conserved property or has arisen independently in different lineages. Details of methods employed to produce the new receptor tree are included in line 727-739. The new phylogenetic tree is shown below and has been incorporated into the revised manuscript (Figure 2, line 206). The description of new phylogenetic tree has also been modified accordingly in the revised manuscript (line 169-183).

References:

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Reviewer #2 (Public review):

Summary:

*The authors show that *A. japonicus* calcitonins (AjCT1 and AjCT2) activate not only the calcitonin/calcitonin-like receptor, but they also activate the two "PDF receptors", ex vivo. They also explore secondary messenger pathways that are recruited following receptor activation. They determine the source of CT1 and CT2 using qPCR and in situ hybridization and finally test the effects of these peptides on tissue contractions, feeding and growth. This study provides solid evidence that CT1 and CT2 act as ligands for calcitonin receptors; however, evidence supporting cross-talk between CT peptides and "PDF receptors" is weak.*

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This is the first study to report pharmacological characterization of CT receptors in an echinoderm. Multiple lines of evidence in cell culture (receptor internalization and

secondary messenger pathways) support this conclusion.

Weaknesses:

*The authors claim that *A. japonicus* CTs activate "PDF" receptors and suggest that this cross-talk is evolutionary ancient since similar phenomenon also exists in the fly *Drosophila melanogaster*. These conclusions are not fully supported. The authors perform phylogenetic analysis to show that the two "PDF" receptors form an independent clade. The bootstrap support is quite low in a lot of instances, especially for the deuterostomian and protostomian PDFR clades which is below 30. With such low support, it is unclear if the clade comprising deuterostomian "PDFR" is in fact PDFRs and not another receptor type whose endogenous ligand (besides CT) remains to be discovered.*

Thank you for the reviewer's comments. In response, we have produced a new phylogenetic analysis using the maximum likelihood method. This was done by Nayeli Escudero Castela and Kite Jones in the Elphick group at QMUL and therefore they have been added as co-authors of this paper. The new phylogenetic tree (Figure 2, line 206) includes broad taxonomic sampling of CT-type receptors and PDF-type receptors. CRH-type receptors, which are also members of the secretin-type GPCR sub-family, have been included as an outgroup to root the tree. In the previous version the much more distantly related vasopressin/oxytocin-type receptors, which are rhodopsin-type GPCRs, were included as an outgroup. Furthermore, VIP-type receptors were also included in the previous tree but these have been omitted from the new tree because VIP receptor orthologs only occur in vertebrates and therefore they are not representative of a bilaterian GPCR family. The new tree shows high bootstrap support for key clades, notably achieving a bootstrap value of 100 for a clade comprising both deuterostomian and protostomian PDF receptors. This provides important evidence that the *A. japonicus* PDF-type receptors characterized in this study (AjPDFR1, AjPDFR2) are co-orthologs of the PDF-type receptor that has been characterized previously in *Drosophila*. Similarly, there is strong bootstrap support (100) for a clade comprising CT/DH31-type receptors and, importantly, the CT-type receptor characterized in this study (AjCTR) is positioned in a branch of this clade that comprises deuterostomian CT-type receptors (with bootstrap support of 100). Details of methods employed to produce the new receptor tree are included in lines 727-739. The new phylogenetic tree is shown below and has been incorporated into the revised manuscript (Figure 2, line 206). The description of new phylogenetic tree has also been modified accordingly in the revised manuscript (line 169-183).

References:

- Bauknecht P, Jékely G. Large-Scale Combinatorial Deorphanization of Platynereis Neuropeptide GPCRs. *Cell reports*, 2015, 12(4), 684–693. doi: 10.1016/j.celrep.2015.06.052.
- Beets I, Zels S, Vandeweyer E, Demeulemeester J, et al. System-wide mapping of peptide-GPCR interactions in *C. elegans*. *Cell reports*, 2023, 42(9), 113058. doi: 10.1016/j.celrep.2023.113058.
- Cardoso J C, Mc Shane J C, Li Z, et al. Revisiting the evolution of Family B1 GPCRs and ligands: Insights from mollusca. *Molecular and cellular endocrinology*, 2024, 586, 112192. doi: 10.1016/j.mce.2024.112192.
- Gorn A H, Lin H Y, Yamin M, et al. Cloning, characterization, and expression of a human calcitonin receptor from an ovarian carcinoma cell line. *The Journal of clinical investigation*, 1992, 90(5), 1726–1735. doi: 10.1172/JCI116046.
- Huang T, Su J, Wang X, et al. Functional Analysis and Tissue-Specific Expression of Calcitonin and CGRP with RAMP-Modulated Receptors CTR and CLR in Chickens. *Animals: an open access journal from MDPI*, 2024, 14(7), 1058. doi: 10.3390/ani14071058.

Johnson E C, Shafer O T, Trigg J S, et al. A novel diuretic hormone receptor in *Drosophila*: evidence for conservation of CGRP signaling. *Journal of Experimental Biology*, 2005, 208(7): 1239-1246. doi: 10.1242/jeb.01529.

McLatchie L M, Fraser N J, Main M J, et al. RAMPs regulate the transport and ligand specificity of the calcitonin-receptor-like receptor. *Nature*, 1998, 393(6683): 333-339. doi: 10.1038/30666.

Schwartz J, Réalis-Doyelle E, Dubos M P, et al. Characterization of an evolutionarily conserved calcitonin signaling system in a lophotrochozoan, the Pacific oyster (*Crassostrea gigas*). *Journal of Experimental Biology*, 2019, 222(13): jeb201319. doi: 10.1242/jeb.201319.

Seikiguchi T, Kuwasako K, Ogasawara M, et al. Evidence for conservation of the calcitonin superfamily and activity-regulating mechanisms in the basal chordate *Branchiostoma floridae*: insights into the molecular and functional evolution in chordates. *Journal of Biological Chemistry*, 2016, 291(5): 2345-2356. doi: 10.1074/jbc.M115.664003.

Reviewer #2 (Recommendations for the authors):

Figure 1C: The bootstrap support is quite low in a lot of instances, especially for the deuterostomian and protostomian PDFR clades which is below 30. With such support, I would be hesitant to label the blue clade as deuterostomian PDFR for two reasons: 1) no members of this clade have been shown to be activated by a PDF-like substance and 2) the current study shows that these receptors are activated by CT-type peptides. Therefore, the phylogenetic analyses do not support the conclusions of this paper. What is the basis for calling these receptors PDFR and not CTR in light of weak phylogenetic support?

Thank you for the reviewer's comments. In response, we have produced a new phylogenetic analysis using the maximum likelihood method. This was done by Nayeli Escudero Castela and Kite Jones in the Elphick group at QMUL and therefore they have been added as co-authors of this paper. The new phylogenetic tree (Figure 2, line 206) includes broad taxonomic sampling of CT-type receptors and PDF-type receptors. CRH-type receptors, which are also members of the secretin-type GPCR sub-family, have been included as an outgroup to root the tree. In the previous version the much more distantly related vasopressin/oxytocin-type receptors, which are rhodopsin-type GPCRs, were included as an outgroup. Furthermore, VIP-type receptors were also included in the previous tree but these have been omitted from the new tree because VIP receptor orthologs only occur in vertebrates and therefore they are not representative of a bilaterian GPCR family. The new tree shows high bootstrap support for key clades, notably achieving a bootstrap value of 100 for a clade comprising both deuterostomian and protostomian PDF receptors. This provides important evidence that the *A. japonicus* PDF-type receptors characterized in this study (AjPDFR1, AjPDFR2) are co-orthologs of the PDF-type receptor that has been characterized previously in *Drosophila*. Similarly, there is strong bootstrap support (100) for a clade comprising CT/DH31-type receptors and, importantly, the CT-type receptor characterized in this study (AjCTR) is positioned in a branch of this clade that comprises deuterostomian CT-type receptors (with bootstrap support of 100). Details of methods employed to produce the new receptor tree are included in lines 727-739. The new phylogenetic tree is shown below and has been incorporated into the revised manuscript (Figure 2, line 206). The description of new phylogenetic tree has also been modified accordingly in the revised manuscript (line 169-183).

We agree with the reviewer that no members of the PDF-type receptor clade in deuterostomes have yet been shown to be activated by a PDF-like substance. That is because the precursors of the PDF-type neuropeptides in echinoderms remain unidentified so far, which precludes clear pharmacological characterization of these receptors within the

deuterostomian PDFR clade. However, the new phylogenetic tree now provides strong support (bootstrap value = 100) for the clade comprising deuterostomian and protostomian PDFRs, confirming the classification of AjPDFR1 and AjPDFR2 as PDF-type receptors.

References:

- Bauknecht P, Jékely G. Large-Scale Combinatorial Deorphanization of Platynereis Neuropeptide GPCRs. *Cell reports*, 2015, 12(4), 684–693. doi: 10.1016/j.celrep.2015.06.052.
- Beets I, Zels S, Vandewyler E, Demeulemeester J, et al. System-wide mapping of peptide-GPCR interactions in *C. elegans*. *Cell reports*, 2023, 42(9), 113058. doi: 10.1016/j.celrep.2023.113058.
- Cardoso J C, Mc Shane J C, Li Z, et al. Revisiting the evolution of Family B1 GPCRs and ligands: Insights from mollusca. *Molecular and cellular endocrinology*, 2024, 586, 112192. doi: 10.1016/j.mce.2024.112192.
- Gorn A H, Lin H Y, Yamin M, et al. Cloning, characterization, and expression of a human calcitonin receptor from an ovarian carcinoma cell line. *The Journal of clinical investigation*, 1992, 90(5), 1726–1735. doi: 10.1172/JCI116046.
- Huang T, Su J, Wang X, et al. Functional Analysis and Tissue-Specific Expression of Calcitonin and CGRP with RAMP-Modulated Receptors CTR and CLR in Chickens. *Animals: an open access journal from MDPI*, 2024, 14(7), 1058. doi: 10.3390/ani14071058.
- Johnson E C, Shafer O T, Trigg J S, et al. A novel diuretic hormone receptor in *Drosophila*: evidence for conservation of CGRP signaling. *Journal of Experimental Biology*, 2005, 208(7): 1239-1246. doi: 10.1242/jeb.01529.
- McLatchie L M, Fraser N J, Main M J, et al. RAMPs regulate the transport and ligand specificity of the calcitonin-receptor-like receptor. *Nature*, 1998, 393(6683): 333-339. doi: 10.1038/30666.
- Schwartz J, Réalis-Doyelle E, Dubos M P, et al. Characterization of an evolutionarily conserved calcitonin signaling system in a lophotrochozoan, the Pacific oyster (*Crassostrea gigas*). *Journal of Experimental Biology*, 2019, 222(13): jeb201319. doi: 10.1242/jeb.201319.
- Sekiguchi T, Kuwasako K, Ogasawara M, et al. Evidence for conservation of the calcitonin superfamily and activity-regulating mechanisms in the basal chordate *Branchiostoma floridae*: insights into the molecular and functional evolution in chordates. *Journal of Biological Chemistry*, 2016, 291(5): 2345-2356. doi: 10.1074/jbc.M115.664003.

The new results following AjCT and AjPDFR2 knockdown are a welcome addition. While this additional evidence supports the claim that AjCT could mediate its effects via AjPDFR2, this evidence does not show that AjCT acts as an endogenous ligand for PDFR in vivo. In combination with the weak phylogenetic analyses, I would recommend the authors to key down their claims that they have functionally characterized a PDFR (in the title and text).

Thank you for your insightful comments and we do understand the reviewer's concern.

Regarding “the weak phylogenetic analyses”, as highlighted above, we have produced a new phylogenetic tree (Fig 2, line 206) that provides strong bootstrap support for the clade comprising deuterostome and protostome PDF-type receptors. For this reason, it is our opinion that inclusion of “pigment-dispersing factor-type receptors” in the title of the paper is appropriate. The details of phylogenetic analysis method were added in line 727-739, and the updated phylogenetic tree has been incorporated into the revised manuscript (Figure 2, line 206). The description of new phylogenetic tree has also been modified accordingly in the revised manuscript (line 169-183). Besides, long-term knockdown of the AjCT precursor and

AjPDFR2 both resulted in identical and significant growth defect phenotypes. And the observation of phenotypic overlap is widely accepted in genetic research as strong evidence for pathway association (Shafer and Taghert, 2009; Van Sinay et al., 2017). This high degree of phenotypic consistency, coupled with our in vitro finding that AjCT2 specifically activates AjPDFR2, strongly supports the conclusion that AjCT2 and AjPDFR2 function within the same signaling pathway in vivo, with AjPDFR2 serving as the key receptor functionally activated by AjCT2.

References:

Shafer, O. T., & Taghert, P. H. (2009). RNA-interference knockdown of *Drosophila* pigment dispersing factor in neuronal subsets: the anatomical basis of a neuropeptide's circadian functions. *PloS one*, 4(12), e8298. doi: 10.1371/journal.pone.0008298.

Van Sinay, E., Mirabeau, O., Depuydt, G., Van Hiel, M. B., Peymen, K., Watteyne, J., Zels, S., Schoofs, L., & Beets, I. (2017). Evolutionarily conserved TRH neuropeptide pathway regulates growth in *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences of the United States of America*, 114(20), E4065–E4074. doi: 10.1073/pnas.1617392114.

Since there is no formal logic defining the use of "type" vs "like" vs "related", I would encourage the authors to use one term (of their choice) to avoid unnecessary confusion. Or another possibility is that these relationships are defined at some point in the manuscript so that it becomes clear to the reader.

Thank you for the reviewer's comments. The "CT-related peptides" has defined in the Introduction (line 54-58). As per your suggestion, we have now defined both "CT-type peptides" and "CT-like peptides" in the Introduction (line 76-79). "CT-type peptides" are characterized by an N-terminal disulphide bridge, whereas "CT-like peptides" (diuretic hormone 31 (DH31)-type peptides) lack this feature. Additionally, in accordance with the definitions, we have corrected these three descriptions in the revised manuscript (line 80, 83, 88 for "CT-type peptides") to ensure consistent and accurate usage of these terms.

"To provide in vivo evidence supporting CT-mediated activation of "PDF" receptors, we conducted the following experiments: Firstly, we confirmed that AjPDFR1 and AjPDFR2 were the functional receptors of AjCT1 and AjCT2 (Figure 2, 3 and 4). Secondly, injection of AjCT2 and siAjCTP1/2-1 in vivo induced corresponding changes in AjPDFR1 and AjPDFR2 expression levels in the intestine (Figure 8C, 9A, 9B and 9C)."

None of these experiments provide direct evidence that CT activates PDFR in vivo. The functional studies are indeed a welcome addition but they cannot discriminate between correlation and causation.

Thank you for the reviewer's insightful comments. We agree that the functional studies do not constitute direct proof that CT's activation of PDFR in vivo. However, we observed identical and significant growth defect phenotypes following long-term knockdown of the AjCT precursor and the AjPDFR2. This high degree of phenotypic congruence, combined with the established in vitro activation of AjPDFR2 by AjCT2, provides strong support for the conclusion that AjCT2 acts as the key endogenous ligand activating the AjPDFR2 signaling pathway in vivo. Importantly, such phenotypic overlap has been widely accepted in genetic research as strong evidence for functional pathway association (Shafer and Taghert, 2009; Van Sinay et al., 2017).

References:

Shafer, O. T., & Taghert, P. H. (2009). RNA-interference knockdown of *Drosophila* pigment dispersing factor in neuronal subsets: the anatomical basis of a neuropeptide's circadian

functions. PloS one, 4(12), e8298. doi: 10.1371/journal.pone.0008298.

Van Sinay, E., Mirabeau, O., Depuydt, G., Van Hiel, M. B., Peymen, K., Watteyne, J., Zels, S., Schoofs, L., & Beets, I. (2017). Evolutionarily conserved TRH neuropeptide pathway regulates growth in *Caenorhabditis elegans*. Proceedings of the National Academy of Sciences of the United States of America, 114(20), E4065–E4074. doi: 10.1073/pnas.1617392114.

<https://doi.org/10.7554/eLife.101799.3.sa0>